DINOv2: Learning Robust Visual Features without Supervision

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Abstract

The recent breakthroughs in natural language processing for model pretraining on large quantities of data have opened the way for similar foundation models in computer vision. These models could greatly simplify the use of images in any system by producing general-purpose visual features, i.e., features that work across image distributions and tasks without finetuning. This work shows that existing pretraining methods, especially self-supervised methods, can produce such features if trained on enough curated data from diverse sources. We revisit existing approaches and combine different techniques to scale our pretraining in terms of data and model size. Most of the technical contributions aim at accelerating and stabilizing the training at scale. In terms of data, we propose an automatic pipeline to build a dedicated, diverse, and curated image dataset instead of uncurated data, as typically done in the self-supervised literature. In terms of models, we train a ViT model (Dosovitskiy et al., 2021) with 1B parameters and distill it into a series of smaller models that surpass the best available general-purpose features, OpenCLIP (Ilharco et al., 2021) on most of the benchmarks at image and pixel levels.

1 Introduction

Learning task-agnostic pretrained representations have become the standard in Natural Language Processing (NLP) (Radford et al., 2019; Raffel et al., 2020; Chowdhery et al., 2022; Hoffmann et al., 2022; Touvron et al., 2023). One can use these features "as they are", i.e., without fine-tuning, and achieve performances on downstream tasks that are significantly better than those produced by task-specific models (Brown et al., 2020). This success has been fueled by pretraining on large quantities of raw text using pretext objectives, such as language modeling (Radford et al., 2017) or word vectors (Devlin et al., 2019), that require no supervision.

Following this paradigm shift in NLP, we expect similar "foundation" models to appear in computer vision (Bommasani et al., 2021). These models should generate visual features that work out of the box on any task, both at the image level, e.g., image classification, and pixel level, e.g., segmentation. Most promising efforts towards these foundation models focus on text-guided pretraining, i.e., using a form of textual supervision to guide the training of the features (Joulin et al., 2016; Mahajan et al., 2018; Radford et al.,

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DINOv2：无需监督学习稳健视觉特征

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摘要

近期自然语言处理领域在大规模数据模型预训练方面的突破，为计算机视觉领域类似的基础模型开辟了道路。这类模型能通过生成通用视觉特征（即无需微调即可跨图像分布与任务工作的特征），极大简化图像在任何系统中的使用。本研究表明，若在足够多源精选数据上进行训练，现有预训练方法（尤其是自监督方法）可产生此类特征。我们重新审视现有方法，结合不同技术从数据和模型规模两个维度扩展预训练。大部分技术贡献聚焦于大规模训练的加速与稳定化：数据方面，我们提出自动构建专用、多样、精选图像数据集的流程，以替代自监督文献中常用的未筛选数据；模型方面，我们训练了包含10亿参数的ViT模型（Dosovitskiy等人，2021），并将其蒸馏为一系列较小模型。这些模型在大多数图像级与像素级基准测试中，超越了当前最佳通用特征模型OpenCLIP（Ilharco等人，2021）。

1 引言

学习任务无关的预训练表示已成为自然语言处理（NLP）领域的标准方法（Radford et al., 2019; Raffel et al., 2020; Chowdhery et al., 2022; Hoffmann et al., 2022; Touvron et al., 2023）。这些特征可以直接“原样使用”，即无需微调，就能在下游任务上取得显著优于特定任务模型的性能（Brown et al., 2020）。这一成功得益于基于大量原始文本的预训练，其采用无需监督的代理目标，例如语言建模（Radford et al., 2017）或词向量（Devlin et al., 2019）。

遵循自然语言处理领域的这一范式转变，我们预期计算机视觉领域也将出现类似的“基础”模型（Bommasani等人，2021）。这些模型应能生成即插即用的视觉特征，适用于图像级别（如图像分类）和像素级别（如分割）的任何任务。目前最有前景的基础模型研究方向集中于文本引导的预训练，即通过某种形式的文本监督来指导特征学习（Joulin等人，2016；Mahajan等人，2018；Radford等人，

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(a) (b) (c) (d)

Figure 1: **Visualization of the first PCA components.** We compute a PCA between the patches of the images from the same column (a, b, c and d) and show their first 3 components. Each component is matched to a different color channel. Same parts are matched between related images despite changes of pose, style or even objects. Background is removed by thresholding the first PCA component.

2021). This form of text-guided pretraining limits the information that can be retained about the image since captions only approximate the rich information in images, and complex pixel-level information may not surface with this supervision. Furthermore, these image encoders require aligned text-image corpora and hence, do not offer the flexibility of their text counterparts, that is, to learn from raw data alone.

An alternative to text-guided pretraining is self-supervised learning (Caron et al., 2018; Chen et al., 2020; He et al., 2022) where features are learned from images alone. These approaches are conceptually closer to pretext tasks such as language modeling and can capture information at the image and pixel level (Caron et al., 2021). Additionally, the features output by self-supervised models have been shown to exhibit various useful properties, and have enabled enabled a wide variety of applications (Amir et al., 2022; Tumanyan et al., 2022; Ofri-Amar et al., 2023; Hamilton et al., 2022). However, despite their potential to learn general-purpose features, most of the advances in self-supervised learning were made in the context of pretraining on a small curated dataset, ImageNet-1k (Russakovsky et al., 2015). Some efforts on scaling these approaches beyond ImageNet-1k have been attempted (Caron et al., 2019; Goyal et al., 2021; 2022a), but they focused on uncurated datasets, which typically lead to a significant drop in the quality of the features. This is explained by the lack of control over the data quality and diversity, which are essential to produce good features.

In this work, we explore if self-supervised learning has the potential to learn general-purpose visual features if pretrained on a large quantity of curated data. We revisit existing discriminative self-supervised approaches that learn features at both the image and patch level, such as iBOT (Zhou et al., 2022a), and we reconsider some of their design choices under the lens of a larger dataset. Most of our technical contributions are tailored toward stabilizing and accelerating discriminative self-supervised learning when scaling in model and data sizes. These improvements make our approach around 2× faster and require 3× less memory than similar discriminative self-supervised methods, allowing us to leverage longer training with larger batch sizes.

Regarding pretraining data, we have built an automatic pipeline to filter and rebalance datasets from an extensive collection of uncurated images. This pipeline is inspired by pipelines used in NLP (Wenzek et al., 2020), where data similarities are used instead of external metadata and do not require manual annotation. A major difficulty when dealing with images in the wild is to rebalance concepts and avoid overfitting on a few dominant modes. In this work, a naive clustering approach works reasonably well to resolve this issue. We gathered a small but diverse corpus of 142M images to validate our approach.

(a) (b) (c) (d)

图1：第一主成分的可视化。我们对同一列（a、b、c和d）图像中的图像块进行主成分分析，并展示其前三个主成分。每个主成分对应不同的颜色通道。尽管姿态、风格甚至物体发生变化，相关图像间的相同部分仍能匹配。通过阈值化第一主成分，背景已被移除。

2021年）。这种文本引导的预训练形式限制了图像信息的保留能力，因为描述文字仅能近似图像中的丰富信息，复杂的像素级信息可能无法通过这种监督方式显现。此外，这些图像编码器需要对齐的文本-图像语料库，因此无法像其文本对应模型那样仅从原始数据中学习，缺乏灵活性。

文本引导预训练的一种替代方案是自监督学习（Caron等人，2018；Chen等人，2020；He等人，2022），其仅从图像中学习特征。这些方法在概念上更接近语言建模等预文本任务，并能捕捉图像和像素级别的信息（Caron等人，2021）。此外，自监督模型输出的特征已被证明具有多种有用特性，并促成了广泛的应用（Amir等人，2022；Tumanyan等人，2022；Ofri-Amar等人，2023；Hamilton等人，2022）。然而，尽管自监督学习具备学习通用特征的潜力，但其大部分进展是在小型精选数据集ImageNet-1k（Russakovsky等人，2015）上进行预训练的背景下取得的。虽然已有一些尝试将这类方法扩展到ImageNet-1k之外（Caron等人，2019；Goyal等人，2021；2022a），但这些工作主要关注未经筛选的数据集，这通常会导致特征质量显著下降。其原因在于缺乏对数据质量和多样性的控制，而这正是生成优质特征的关键。

在本研究中，我们探讨了自监督学习是否具备通过大量精选数据预训练来学习通用视觉特征的潜力。我们重新审视了现有的判别式自监督方法（例如iBOT（Zhou et al., 2022a）），这些方法可在图像和局部块级别学习特征，并在更大数据集的视角下重新考量其部分设计选择。我们的技术贡献主要聚焦于在扩展模型与数据规模时，提升判别式自监督学习的稳定性与训练效率。这些改进使我们的方法比同类判别式自监督方法训练速度提升约2×倍、内存消耗降低3×倍，从而能够支持更大批量的长时间训练。

关于预训练数据，我们构建了一个自动化流程，用于从大量未经整理的图像中筛选和重新平衡数据集。该流程受自然语言处理中采用的流程（Wenzek等人，2020年）启发，其中使用数据相似性而非外部元数据，且无需人工标注。处理真实场景图像时的一个主要难点在于重新平衡概念并避免对少数主导模式过拟合。在本工作中，一种简单的聚类方法能较好地解决这一问题。我们收集了一个小而多样的142M图像语料库以验证我们的方法。

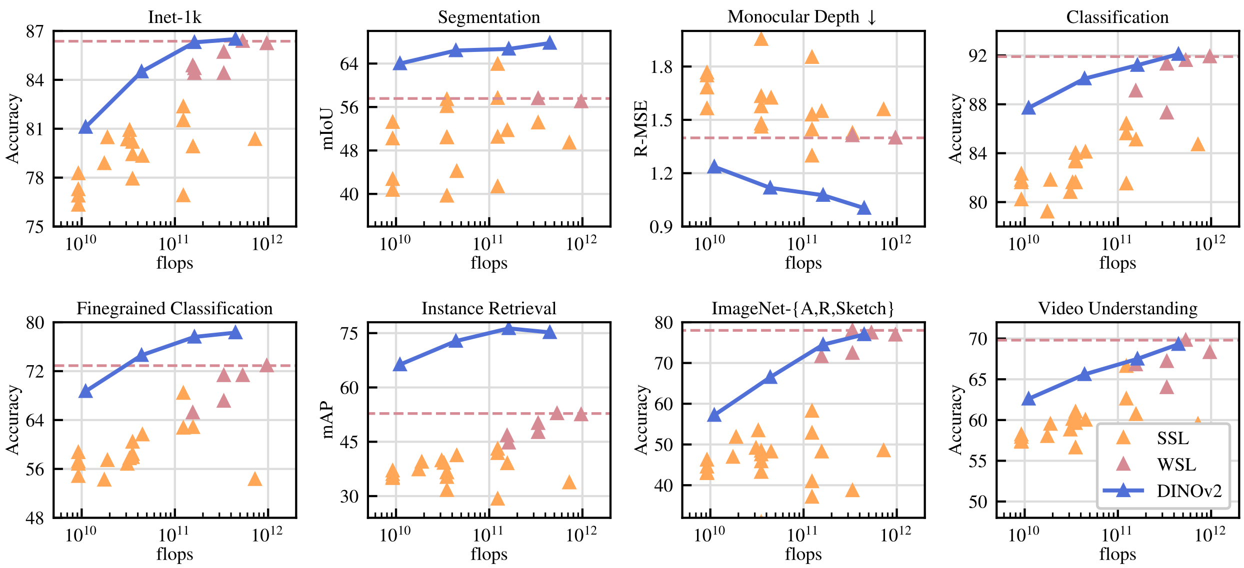


Figure 2: **Evolution of performance when scaling in parameters.** We show performance on eight types of vision tasks, as presented in Sec. 7, and average metrics with each type. Features are extracted from our self-supervised encoders, DINOv2 (dark blue), and we compare them with self-supervised methods (pale orange), as well as weakly-supervised methods (dark pink). We report the best-performing weakly-supervised model's performance as a dashed horizontal line. Our family of models drastically improves over the previous state of the art in self-supervised learning and reaches performance comparable with weakly-supervised features. See Sec. 7 for a detailed analysis.

Finally, we provide a variety of pretrained visual models, called DINOv2, trained with different Vision Transformers (ViT) (Dosovitskiy et al., 2016) architectures on our data. We release all the models and the code to retrain DINOv2 on any data. We validate the quality of DINOv2 on various computer vision benchmarks at both image and pixel levels as we scale them, as summarized in Fig. 2. We conclude that self-supervised pretraining alone is a good candidate for learning transferable frozen features that are competitive with the best openly available weakly-supervised models.

2 Related Work

Intra-image self-supervised training. A first family of self-supervised methods focuses on pretext tasks built from the image, i.e., extracting a signal from the image to be predicted from the rest of the image. This idea has become prevalent with the work of Doersch et al. (2015), where they train by predicting the context of a given patch. Many other pretext tasks were introduced based on, for example, re-colorizing images (Zhang et al., 2016), predicting transformations (Gidaris et al., 2018), inpainting (Pathak et al., 2016) or patch re-ordering (Noroozi & Favaro, 2016; Misra & Maaten, 2020). Recently, the emergence of patch-based architectures, like ViTs, has led to a revisit of inpainting for pre-training (He et al., 2022; Bao et al., 2021; El-Nouby et al., 2021), potentially in feature space (Assran et al., 2023; Baevski et al., 2022). Of particular interest, He et al. (2022) show that a masked auto-encoder (MAE) learns features that provide substantial improvements when finetuned on downstream tasks. This property of MAEs has been further validated on video (Tong et al., 2022), audio (Xu et al., 2022), and across other modalities (Girdhar et al., 2023). However, their features require supervised finetuning, while our features perform well out of the box.

Discriminative self-supervised learning. The second line of work, closer to ours, is using discriminative signals between images or groups of images to learn features. This family of methods has roots in early deep learning work (Hadsell et al., 2006) but became popular with the emergence of instance classification methods (Dosovitskiy et al., 2016; Bojanowski & Joulin, 2017; Wu et al., 2018). Several improvements

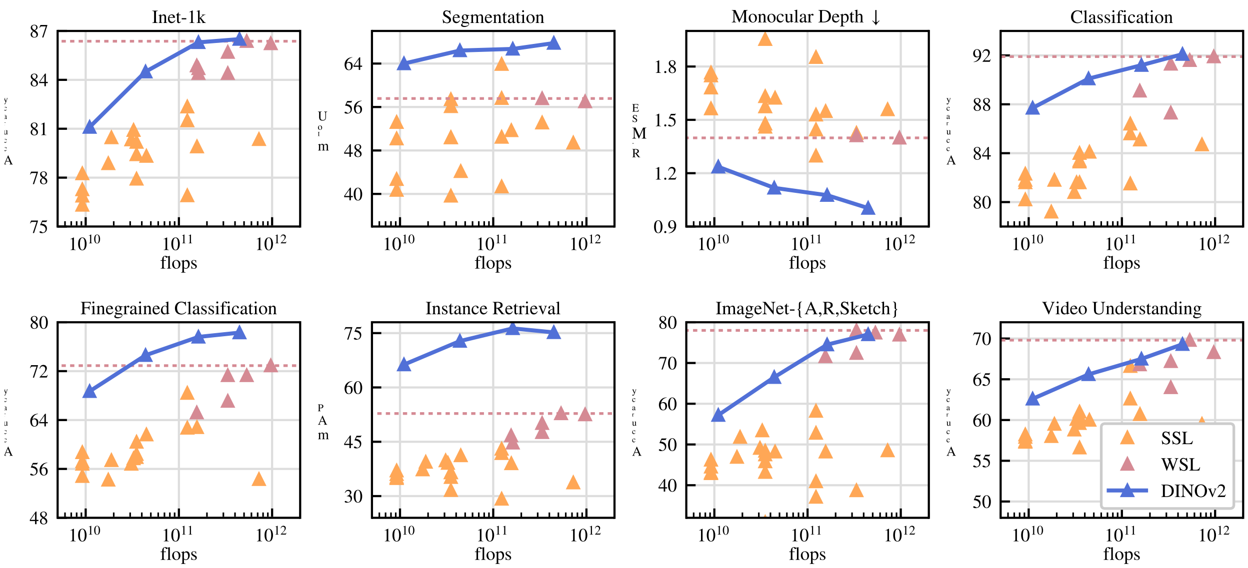


图2：参数量扩展时的性能演变。我们在八类视觉任务上展示性能表现（如第7节所述），并给出每类任务的平均指标。特征提取自我们的自监督编码器DINOv2（深蓝色），同时与自监督方法（浅橙色）及弱监督方法（深粉色）进行对比。最佳弱监督模型的性能以虚线水平线标示。我们的模型家族显著超越了自监督学习的先前最优水平，并达到了与弱监督特征相媲美的性能。详细分析参见第7节。

最后，我们提供了一系列预训练的视觉模型，称为DINOv2，这些模型采用不同的视觉变换器（ViT）架构（Dosovitskiy等人，2016年）在我们的数据上进行训练。我们发布了所有模型以及用于在任何数据上重新训练DINOv2的代码。随着模型规模的扩大，我们在多个计算机视觉基准测试中从图像和像素层面验证了DINOv2的质量，如图2所示。我们得出结论，仅通过自监督预训练就能学习到具有可迁移性的冻结特征，这些特征与当前公开可用的最佳弱监督模型相比具有竞争力。

2 相关工作

图像内自监督训练。第一类自监督方法专注于基于图像构建的预训练任务，即从图像中提取信号，使其能够从图像的其余部分进行预测。这一思想随着Doersch等人（2015）的研究而变得流行，他们通过预测给定图像块的上下文进行训练。随后出现了许多其他预训练任务，例如图像重新着色（Zhang等人，2016）、预测变换（Gidaris等人，2018）、图像修复（Pathak等人，2016）或图像块重排序（Noroozi & Favaro, 2016; Misra & Maaten, 2020）。近年来，随着基于图像块的架构（如ViTs）的出现，图像修复预训练方法被重新审视（He等人，2022；Bao等人，2021；El-Nouby等人，2021），并可能扩展到特征空间（Assran等人，2023；Baevski等人，2022）。特别值得注意的是，He等人（2022）表明，掩码自编码器（MAE）学习的特征在下游任务微调时能带来显著性能提升。MAE的这一特性已在视频（Tong等人，2022）、音频（Xu等人，2022）及其他模态（Girdhar等人，2023）中得到进一步验证。然而，MAE的特征需要监督微调才能发挥作用，而我们的特征无需调整即可取得良好性能。

判别式自监督学习。第二条研究路线与我们的工作更为接近，即利用图像或图像组之间的判别信号来学习特征。这类方法源于早期深度学习研究（Hadsell等人，2006），但随着实例分类方法的出现而流行起来（Dosovitskiy等人，2016；Bojanowski & Joulin，2017；Wu等人，2018）。后续多项改进

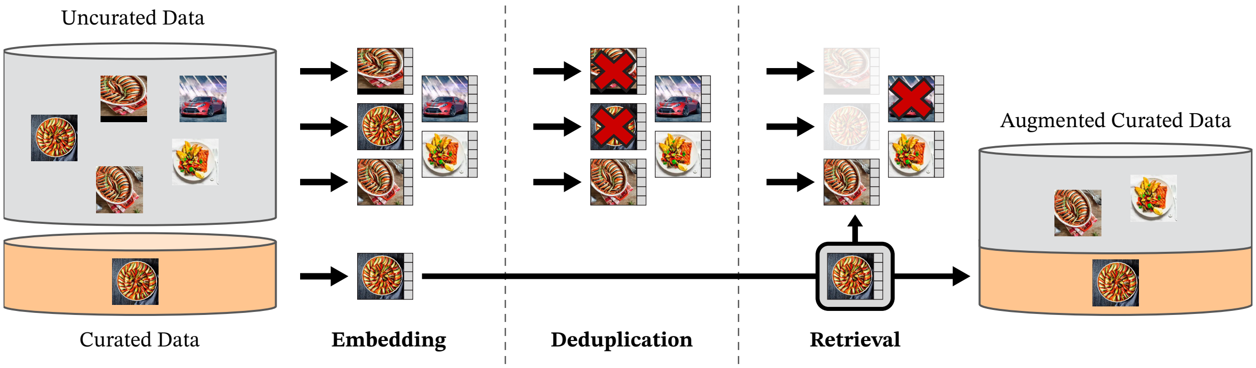


Figure 3: **Overview of our data processing pipeline.** Images from curated and uncurated data sources are first mapped to embeddings. Uncurated images are then deduplicated before being matched to curated images. The resulting combination augments the initial dataset through a self-supervised retrieval system.

were made based either on instance-level objectives (Hénaff et al., 2019; He et al., 2020; Chen & He, 2021; Chen et al., 2020; Grill et al., 2020; Caron et al., 2021) or clustering (Caron et al., 2018; Asano et al., 2020; Caron et al., 2020). These methods provide performant frozen features on standard benchmarks like ImageNet (Russakovsky et al., 2015), but they are hard to scale to larger model sizes (Chen et al., 2021). In this work, we revisit the training of these approaches in the context of large pretraining datasets and models. In particular, we build on top of Zhou et al. (2022a) that we find particularly suited for scaling.

Scaling self-supervised pretraining. A growing body of work has focused on the scaling abilities of self-supervised learning in terms of data and model size (Caron et al., 2019; Goyal et al., 2019; Tian et al., 2021; Goyal et al., 2022a). Most of these works use large quantities of uncurated data to train models without supervision. They show evidence that discriminative methods scale with data, but because of the poor quality of the pretraining data, most of the results are obtained by finetuning the features. Of particular interest, Goyal et al. (2021) have also shown that these methods benefit from scaling in model size given enough pretrained data. This line of work questions the ability of self-supervised methods to work on any data while we focus on producing the best pretrained encoders.

Automatic data curation. Our dataset construction borrows from the image retrieval community (Weinzaepfel et al., 2021; Radenović et al., 2018b; Berman et al., 2019; Douze et al., 2009; Tolias et al., 2016; Revaud et al., 2019). In particular, the use of retrieval to augment the training set has been studied in the context of semi-supervised learning (Yalniz et al., 2019). Similarly, others have used hashtags or other metadata (Mahajan et al., 2018; Radford et al., 2021) or pretrained vision encoders (Schuhmann et al., 2021; 2022) to filter uncurated datasets. Unlike these works, we use no pretrained encoders, metadata nor supervision to filter images and leverage visual similarity between images. Our approach is inspired by text curation pipelines (Wenzek et al., 2020), where a language model is trained on Wikipedia to score texts extracted from an uncurated source.

3 Data Processing

We assemble our curated LVD-142M dataset by retrieving, from a large pool of uncurated data, images that are close to those in several curated datasets. We describe below the main components in our data pipeline including the curated/uncurated data sources, the image deduplication step and the retrieval system. Our pipeline does not require any metadata or text and directly works with images, as shown in Fig. 3. We refer the reader to appendix A for more details on our approach.

Data sources. Our selection of curated datasets is detailed in the appendix (Table 15) and contains ImageNet-22k, the train split of ImageNet-1k, Google Landmarks and several fine-grained datasets. For the

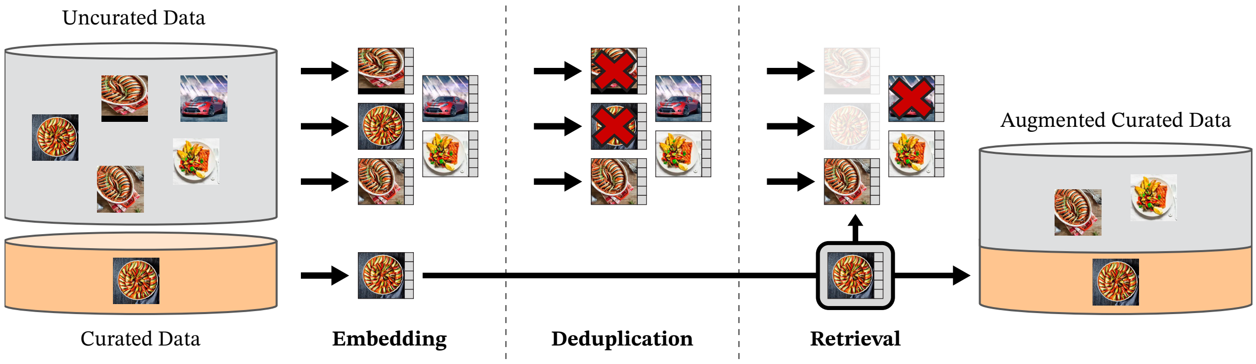


图3：我们的数据处理流程概览。来自精选与非精选数据源的图像首先被映射为嵌入向量。随后，非精选图像在匹配精选图像前会进行去重处理。通过自监督检索系统，最终组合对初始数据集进行了增强。

这些方法要么基于实例级目标（Hénaff等人，2019；He等人，2020；Chen & He，2021；Chen等人，2020；Grill等人，2020；Caron等人，2021），要么基于聚类（Caron等人，2018；Asano等人，2020；Caron等人，2020）进行设计。它们在ImageNet（Russakovsky等人，2015）等标准基准测试中提供了高性能的冻结特征，但难以扩展到更大的模型规模（Chen等人，2021）。在本工作中，我们在大规模预训练数据集和模型的背景下重新审视这些方法的训练过程。特别地，我们基于Zhou等人（2022a）的研究进行拓展，发现其方法尤其适合规模化扩展。

扩展自监督预训练的规模。越来越多的研究关注自监督学习在数据和模型规模上的扩展能力（Caron等人，2019；Goyal等人，2019；Tian等人，2021；Goyal等人，2022a）。这些研究大多使用大量未经筛选的数据进行无监督模型训练。它们证明判别式方法能随数据规模扩展，但由于预训练数据质量较低，多数结果需通过微调特征获得。特别值得注意的是，Goyal等人（2021）还证明了在拥有足够预训练数据的前提下，这些方法能通过扩大模型规模获得提升。这类研究对自监督方法能否适用于任意数据提出了质疑，而我们的研究重点在于生成最优的预训练编码器。

自动数据整理。我们的数据集构建借鉴了图像检索领域的研究（Weinzaepfel等人，2021；Radenovi　等人，2018b；Berman等人，2019；Douze等人，2009；Tolias等人，2016；Revaud等人，2019）。特别是在半监督学习背景下，利用检索技术增强训练集的方法已被探索（Yalniz等人，2019）。类似地，其他研究通过使用标签或其他元数据（Mahajan等人，2018；Radford等人，2021）或预训练视觉编码器（Schuhmann等人，2021；2022）来筛选未经整理的数据集。与这些工作不同，我们既不使用预训练编码器、元数据，也不依赖监督来筛选图像，而是利用图像间的视觉相似性。我们的方法受到文本整理流程的启发（Wenzek等人，2020），该流程通过在维基百科上训练语言模型，对从未经整理的来源提取的文本进行评分。

3 数据处理

我们通过从大量未筛选数据中检索与多个精选数据集中的图像相近的图片，构建了精心整理的LVD-142M数据集。下文将介绍我们数据流程中的主要组成部分，包括精选/未筛选数据来源、图像去重步骤以及检索系统。如图3所示，我们的流程无需任何元数据或文本，可直接处理图像。关于方法的更多细节，请参阅附录A。

数据来源。我们在附录（表15）中详细介绍了精选数据集的选择，其中包含ImageNet-22k、ImageNet-1k的训练集、Google Landmarks以及多个细粒度数据集。对于

uncurated data source, we collect a raw unfiltered dataset of images from a publicly available repository of crawled web data. From each web page in the repository, we extract URL links of images from `<img>` tags. We discard URLs that are unsafe or restricted by domains, and post-process the downloaded images (PCA hash deduplication, NSFW filtering, and blurring identifiable faces). This results in 1.2B unique images.

Deduplication. We apply the copy detection pipeline of Pizzi et al. (2022) to the uncurated data and remove near-duplicate images. This reduces redundancy and increases diversity among images. We also remove near-duplicates of images contained in the test or validation set of any benchmark used in this work.

Self-supervised image retrieval. We build our curated pretraining dataset by retrieving images from our uncurated data source that are close to images in our curated sources. In order to do this, we first compute an image embedding using a self-supervised ViT-H/16 network pretrained on ImageNet-22k, and use cosine-similarity as a distance measure between images. Then, we perform k-means clustering of the uncurated data. Given a query dataset for retrieval, if it is large enough we retrieve N (typically 4) nearest neighbors for each query image. If it is small, we sample M images from the cluster corresponding to each query image. Although visual inspection seemed to indicate good retrieval quality for N much larger than 4, this leads to more collisions (images that are nearest-neighbor retrievals of multiple queries). We choose $N = 4$ as it provides a good tradeoff in that sense.

Implementation Details. The deduplication and retrieval stages of our pipeline rely on the Faiss library (Johnson et al., 2019) to efficiently index and compute batch searches of nearest embeddings. In particular, we heavily leverage its support for GPU-accelerated indices, using inverted file indices with product quantization codes (Jegou et al., 2010). The whole processing is distributed on a compute cluster of 20 nodes equipped with 8 V100-32GB GPUs and takes less than two days to produce the LVD-142M dataset.

4 Discriminative Self-supervised Pre-training

We learn our features with a discriminative self-supervised method that can be seen as a combination of DINO and iBOT losses with the centering of SwAV (Caron et al., 2020). We also add a regularizer to spread features and a short high-resolution training phase. We rapidly introduce each of these approaches, but more details can be found in the related papers, or in our open-sourced code.

- **Image-level objective (Caron et al., 2021).** We consider the cross-entropy loss between the features extracted from a student and a teacher network. Both features are coming from the class token of a ViT, obtained from different crops of the same image. We pass the student class token through the student DINO head. This head is an MLP model outputting a vector of scores, that we call "prototype scores". We then apply a softmax to obtain p_s. Similarly, we apply the teacher DINO head to the teacher class token to obtain teacher prototype scores. We then apply a softmax followed by a centering with moving average (or a Sinkhorn-Knopp centering as detailed thereafter) to obtain p_t. The DINO loss term corresponds to:

$$\mathcal{L}_{DINO} = -\sum p_t \log p_s$$

  We learn the parameters of the student and build the teacher head with an exponential moving average of past iterates (He et al., 2020).

- **Patch-level objective (Zhou et al., 2022a).** We randomly mask some of the input patches given to the student, but not to the teacher. We then apply the student iBOT head to the student mask tokens. Similarly, we apply the teacher iBOT head to the (visible) teacher patch tokens corresponding to the ones masked in the student. We then apply the softmax and centering steps as above, and obtain the iBOT loss term:

$$\mathcal{L}_{iBOT} = -\sum_i p_{ti} \log p_{si}$$

在未经筛选的数据源中，我们从公开可用的网络爬取数据存储库中收集了一个原始未过滤的图像数据集。从存储库的每个网页中，我们通过<img>标签提取图像的URL链接。我们丢弃了不安全或受域名限制的URL，并对下载的图像进行了后处理（PCA哈希去重、NSFW过滤以及模糊化可识别人脸）。最终得到了12亿张独特图像。

去重。我们对未经筛选的数据应用Pizzi等人（2022）的复制检测流程，移除近似重复图像。这减少了冗余并增加了图像间的多样性。同时，我们还移除了与本工作中使用的任何基准测试集或验证集中图像高度相似的重复图像。

自监督图像检索。我们通过从非精选数据源中检索与精选数据源中图像相近的图像，构建了精选预训练数据集。为此，我们首先使用在ImageNet-22k上预训练的自监督ViT-H/16网络计算图像嵌入向量，并采用余弦相似度作为图像间距离度量。随后，我们对非精选数据进行k均值聚类。对于给定的检索查询数据集，若其规模足够大，我们为每张查询图像检索N (通常为4)个最近邻图像；若规模较小，则从对应每张查询图像的聚类中采样M张图像。尽管视觉检查显示即使N远大于4时检索质量仍表现良好，但这会导致更多碰撞现象（即同一图像被多个查询作为最近邻检索结果）。我们选择N =4作为平衡点，因其在该意义上提供了较好的权衡。

实现细节。我们流程中的去重和检索阶段依赖于Faiss库（Johnson等人，2019），以高效索引并批量计算最近邻嵌入的搜索。具体而言，我们充分利用其对GPU加速索引的支持，采用基于乘积量化码的倒排文件索引（Jegou等人，2010）。整个处理过程分布在由20个节点组成的计算集群上，每个节点配备8块V100-32GB GPU，生成LVD-142M数据集耗时不足两天。

4 判别式自监督预训练

我们采用一种判别性自监督方法学习特征，该方法可视为结合了DINO与iBOT损失函数，并引入了SwAV（Caron等人，2020）的中心化机制。我们还添加了分散特征的正则化器以及一个简短的高分辨率训练阶段。以下将简要介绍这些方法，更多细节可参阅相关论文或我们开源的代码。

- 图像级目标（Caron等人，2021）。我们考虑从学生网络和教师网络提取的特征之间的交叉熵损失。这两个特征均来自ViT的类别标记，通过同一图像的不同裁剪区域获得。我们将学生类别标记输入学生DINO头部。该头部是一个多层感知机模型，输出一组得分向量，我们称之为“原型得分”。随后应用softmax函数得到p_s。类似地，我们将教师DINO头部应用于教师类别标记以获得教师原型得分，接着应用softmax函数并进行移动平均中心化处理（或后续详述的Sinkhorn-Knopp中心化方法）得到p_t。DINO损失项对应为：

$$\mathcal{L}_{DINO} = -\sum p_t \log p_s$$

  我们学习学生模型的参数，并利用过去迭代的指数移动平均构建教师头（He等人，2020年）。

- 块级目标（Zhou等人，2022a）。我们随机遮蔽输入给学生的部分图像块，但不遮蔽教师的输入。随后，我们将学生的iBOT头部应用于学生被遮蔽的标记。类似地，我们将教师的iBOT头部应用于教师图像块标记中与学生被遮蔽块相对应的（可见）标记。接着，我们按上述方法进行softmax和中心化步骤，并得到iBOT损失项：

$$\mathcal{L}_{iBOT} = -\sum_i p_{ti} \log p_{si}$$

, where i are patch indices for masked tokens. Similarly to above, we learn the parameters of the student, and build the teacher head through exponential moving average.

- **Untying head weights between both objectives.** Both the DINO and the iBOT loss use a learnable MLP projection head. It is applied to the output tokens and the loss is compute atop. In Zhou et al. (2022a), an ablation study shows that sharing parameters between the DINO and iBOT heads leads to better performance. At scale, we observed that the opposite is true, and we therefore use two separate heads in all our experiments.

- **Sinkhorn-Knopp centering (Caron et al., 2020).** Ruan et al. (2023) recommend to replace the teacher softmax-centering step of DINO and iBot by the Sinkhorn-Knopp (SK) batch normalization of SwAV (Caron et al., 2020). We run the Sinkhorn-Knopp algorithm steps for 3 iterations. For the student, we apply the softmax normalization.

- **KoLeo regularizer (Sablayrolles et al., 2019).** The KoLeo regularizer derives from the Kozachenko-Leonenko differential entropy estimator (see Beirlant et al. (1997); Delattre & Fournier (2017)) and encourages a uniform span of the features within a batch. Given a set of n vectors $(x_1, \ldots, x_n)$, it is defined as
$$\mathcal{L}_{\text{koleo}} = -\frac{1}{n} \sum_{i=1}^{n} \log(d_{n,i}),$$
where $d_{n,i} = \min_{j \neq i} \|x_i - x_j\|$ is the minimum distance between x_i and any other point within the batch. We also ℓ_2-normalize the features before computing this regularizer.

- **Adapting the resolution (Touvron et al., 2019).** Increasing image resolution is key to pixel-level downstream tasks such as segmentation or detection, where small objects disappear at low resolutions. However, training at high resolution is time and memory demanding, and instead, we increase the resolution of images to 518×518 during a short period at the end of pretraining. This is also similar to UniViT training from Likhomanenko et al. (2021) and FlexiViT training from Beyer et al. (2023).

5 Efficient implementation

We consider several improvements to train models at a larger scale. We train models on A100 GPUs using PyTorch 2.0. The code and pretrained models are made available under Apache 2.0 license [1]. The details of our models are in the appendix, Table 17. With the same hardware, compared to the iBOT implementation, the DINOv2 code runs around 2× faster using only 1/3 of the memory.

Fast and memory-efficient attention. We implemented our own version of FlashAttention (Dao et al., 2022) to improve memory usage and speed on the self-attention layers. Our version is on par with or better than the original on all cases considered, while covering more use-cases and hardware. Due to the GPU hardware specifics, the efficiency is best when the embedding dimension per head is a multiple of 64, and the matrix operations are even better when the full embedding dimension is a multiple of 256. As a consequence, our ViT-g architecture slightly differs from the architecture proposed by Zhai et al. (2022) in order to maximize compute efficiency, and we use an embedding dimension of 1536 with 24 heads (64 dim/head), rather than 1408 with 16 heads (88 dim/head). Our experiments did not show significant differences in final accuracy, and our ViT-g backbone counts 1.1B parameters.

Sequence packing. The DINO algorithm requires forwarding both large crops (at resolution 224) and small crops (resolution 98). When split into patches, these two groups are represented by token sequences of different lengths and cannot be forwarded together. In order to accelerate training, we use a trick called "sequence packing," which originates from NLP (Krell et al., 2022). The idea is simple: we concatenate the

[1] `https://github.com/facebookresearch/dinov2`

其中i是被遮蔽标记的补丁索引。与上述类似，我们学习学生模型的参数，并通过指数移动平均构建教师头。

- 解绑两个目标之间的头部权重。DINO和iBOT损失都使用可学习的MLP投影头，该投影头应用于输出标记，并在其之上计算损失。在Zhou等人（2022a）的研究中，消融实验表明共享DINO和iBOT头部的参数能带来更好的性能。但在大规模实验中，我们观察到相反的情况，因此我们在所有实验中使用两个独立的头部。

- Sinkhorn-Knopp中心化（Caron等人，2020年）。Ruan等人（2023年）建议将DINO和iBot中的教师softmax中心化步骤替换为SwAV（Caron等人，2020年）的Sinkhorn-Knopp（SK）批量归一化。我们运行Sinkhorn-Knopp算法3次迭代。对于学生模型，我们采用softmax归一化。

- KoLeo正则化器（Sablayrolles等人，2019）。该正则化器源自Kozachenko-Leonenko微分熵估计器（参见Beirlant等人（1997）；Delattre & Fournier（2017）），并鼓励批次内特征均匀分布。给定一组n向量$(x_1, \ldots, x_n)$，其定义为

$$\mathcal{L}_{\text{koleo}} = -\frac{1}{n}\sum_{i=1}^{n}\log(d_{n,i}),$$

其中 $d_{n,i} = \min_{j\neq i}\|x_i - x_j\|$ 是 x_i 与批次内任意其他点之间的最小距离。在计算该正则化项前，我们还会对特征进行 ℓ_2 归一化处理。

- 调整分辨率（Touvron等人，2019年）。提高图像分辨率对于像素级下游任务（如分割或检测）至关重要，因为在低分辨率下小物体会消失。然而，在高分辨率下训练对时间和内存要求较高，因此我们改为在预训练结束前的短时间内将图像分辨率提升至518 $\times$ 518。这也类似于Likhomanenko等人（2021年）的UniViT训练和Beyer等人（2023年）的FlexiViT训练。

5 高效实现

我们考虑了几项改进，以更大规模地训练模型。我们使用PyTorch 2.0在A100 GPU上训练模型。代码和预训练模型均基于Apache 2.0许可证[1]公开提供。模型详细参数见附录表17。在相同硬件条件下，与iBOT实现相比，DINOv2代码运行速度提升约2$\times$倍，且仅占用1/3的内存。

快速且内存高效的注意力机制。我们实现了自研版本的FlashAttention（Dao等人，2022年），以提升自注意力层的内存使用效率和运行速度。在所有测试场景中，我们的版本均达到或超越了原始版本性能，同时覆盖了更多使用场景和硬件类型。受GPU硬件特性影响，当每个注意力头的嵌入维度为64的倍数时效率最优，而整体嵌入维度为256的倍数时矩阵运算效率更佳。因此，为最大化计算效率，我们的ViT-g架构与Zhai等人（2022年）提出的方案略有不同：采用1536嵌入维度配合24个注意力头（每头64维），而非原方案的1408维度配合16个注意力头（每头88维）。实验表明两种方案在最终准确率上无显著差异，我们的ViT-g骨干网络共包含11亿参数。

序列打包。DINO算法需要同时处理大尺寸裁剪（分辨率为224）和小尺寸裁剪（分辨率为98）。当分割为图像块时，这两组数据会形成不同长度的标记序列，无法同时进行前向传播。为了加速训练，我们采用了一种称为“序列打包”的技巧，该技巧源自自然语言处理领域（Krell等人，2022）。其思路很简单：我们将

[1]`https://github.com/facebookresearch/dinov2`

sequences we must forward through the transformers into a single long sequence. We pass this sequence through the transformer blocks as usual. However, a block-diagonal mask is applied to the self-attention matrix in attention layers, preventing attention between different sequences. This way, the forward is strictly equivalent to forwarding each sequence separately. This trick gives us significant compute efficiency gains compared to using separate forward and backward passes, as in prior implementations. The lower-level components of our setup are available in the xFormers library[2] (Lefaudeux et al. (2022)).

Efficient stochastic depth. We implement an improved version of stochastic depth (Huang et al., 2016) that skips the computation of the dropped residuals rather than masking the result. This saves memory and compute in proportion approximately equal to the drop rate, thanks to specific fused kernels. With high drop rates ($d = 40\%$ in this work), this allows a drastic improvement in compute efficiency and memory usage. The implementation consists of randomly shuffling the B samples over the batch dimension, and slicing the first $(1 - d) \times B$ samples for the computations in the block.

Fully-Sharded Data Parallel (FSDP). Minimizing our objective with the AdamW optimizer requires 4 model replicas in float32 precision – student, teacher, optimizer first moments, optimizer second moments. This sums to 16 GB of memory for a billion-parameter model such as our ViT-g. In order to reduce this memory footprint per GPU, we split the model replicas across GPUs, i.e., sharding 16 GB across GPUs using the PyTorch implementation of FSDP. Consequently, the model size is not bounded by the memory of a single GPU but by the total sum of GPU memory across compute nodes. The Pytorch implementation of FSDP brings a second advantage, which is to save on the cross-GPU communication costs: the weight shards are stored in float32 precision as required by the optimizer, but broadcasting weights and reducing gradients is done in float16 precision for the backbone (MLP heads gradients are reduced in float32 to avoid training instabilities). This leads to approximately 50% reduction in communication costs compared to the float32 gradient all-reduce operation used in DistributedDataParallel (DDP), which is used in other self-supervised pretraining methods (Caron et al., 2021; Zhou et al., 2022a). As a consequence, the training procedure scales more efficiently than DDP with float16 autocast when scaling the number of GPU nodes. Overall, Pytorch-FSDP mixed-precision is superior to DDP with autocast in virtually all cases we encountered.

Model distillation. Most of our technical improvements to the training loop aim at improving the training of large models over large quantities of data. For smaller models, we distill them from our largest model, the ViT-g, instead of training them from scratch. Knowledge distillation (Hinton et al., 2014) aims at reproducing the output of a large model with a smaller model by minimizing some distance between both outputs for a set of given inputs. Since our objective function is a form of distillation from the teacher network to the student network, we leverage the same training loop with a few exceptions: we use a larger model as a frozen teacher, keep a spare EMA of the student that we use as our final model, remove the masking and stochastic depth, and, apply the iBOT loss on the two global crops. In our ablations, we observe that this approach achieves better performance than training from scratch, even for a ViT-L. Our distillation method ends up close to the one described by Duval et al. (2023), except we do not modify the loss terms for distillation and evaluate the EMA of the student.

6 Ablation Studies

We present a set of ablations to empirically validate different components of our pipeline: the technical modifications described in Sec. 4, the pretraining data and the impact of model distillation. We consider various downstream tasks that are described in Sec. 7.

6.1 Improved Training Recipe

Our approach improves over the iBOT method by combining it with several existing components described in Sec. 4. To evaluate their importance, we train multiple models where we successively add components to a baseline iBOT model. We report the Top-1 accuracy on the validation set of ImageNet-1k with a k-NN

[2] `https://github.com/facebookresearch/xformers`

我们必须通过变换器前向传播的序列合并成一个长序列。我们像往常一样将这个序列通过变换器块。然而，在注意力层的自注意力矩阵上应用了块对角掩码，以防止不同序列之间的注意力交互。这样一来，前向传播严格等同于分别处理每个序列。与先前实现中使用的单独前向和反向传播相比，这种技巧为我们带来了显著的计算效率提升。我们设置中的底层组件可在xFormers库[2]（Lefaudeux等人（2022））中找到。

高效随机深度。我们实现了一种改进版的随机深度（Huang等人，2016），该方法跳过被丢弃残差的计算，而非对结果进行掩码处理。借助特定的融合内核，这能按丢弃率大致成比例地节省内存和计算量。在高丢弃率（本工作中$d=$为40%）下，该方法可大幅提升计算效率和内存使用率。具体实现方式为：在批次维度上随机打乱B个样本，并截取前$(1-d)\times B$个样本用于块内计算。

全分片数据并行（FSDP）。使用AdamW优化器最小化我们的目标需要4个float32精度的模型副本——学生模型、教师模型、优化器一阶矩和优化器二阶矩。对于像我们的ViT-g这样的十亿参数模型，这总计需要16 GB内存。为了降低每个GPU的内存占用，我们将模型副本拆分到多个GPU上，即使用PyTorch实现的FSDP将16 GB内存分片到多个GPU上。因此，模型大小不受单个GPU内存的限制，而是受计算节点间GPU内存总和的限制。PyTorch实现的FSDP还带来第二个优势：节省跨GPU通信成本——权重分片按优化器要求以float32精度存储，但主干网络的权重广播和梯度归约以float16精度执行（MLP头梯度归约使用float32以避免训练不稳定）。与分布式数据并行（DDP）中使用的float32梯度全归约操作相比，这使通信成本降低约50%（其他自监督预训练方法采用DDP，如Caron等人2021年；Zhou等人2022a年）。因此，在扩展GPU节点数量时，该训练流程比使用float16自动混合精度的DDP扩展效率更高。总体而言，在我们遇到的所有实际场景中，PyTorch-FSDP混合精度方案均优于自动混合精度的DDP。

模型蒸馏。我们对训练循环的大部分技术改进旨在提升大模型在大量数据上的训练效果。对于较小模型，我们选择从最大模型ViT-g进行蒸馏，而非从头开始训练。知识蒸馏（Hinton等人，2014）的核心目标是通过最小化给定输入集上大模型与小模型输出间的某种距离，使小模型复现大模型的输出。由于我们的目标函数本质上是教师网络向学生网络的知识蒸馏形式，我们沿用了相同的训练流程，仅作几点调整：使用更大的冻结模型作为教师网络，为学生网络维护一个额外的指数移动平均版本作为最终模型，移除掩码机制和随机深度，并在两个全局裁剪图像上应用iBOT损失。在消融实验中，我们发现即使对于ViT-L模型，该方法也比从头训练获得更优性能。我们的蒸馏方法最终与Duval等人（2023）所述方案相近，区别在于我们未修改蒸馏损失项，且评估的是学生网络的指数移动平均版本。

6 消融研究

我们进行了一系列消融实验，以实证验证我们流程中的不同组成部分：第4节中描述的技术改进、预训练数据以及模型蒸馏的影响。我们考虑了第7节中描述的各种下游任务。

6.1 改进的训练方案

我们的方法通过结合第4节中描述的几种现有组件，改进了iBOT方法。为了评估这些组件的重要性，我们训练了多个模型，逐步向基线iBOT模型添加组件。我们报告了在ImageNet-1k验证集上使用k-NN方法得到的Top-1准确率。

[2]https://github.com/facebookresearch/xformers

	INet-1k k-NN	INet-1k linear
iBOT	72.9	82.3
+(our reproduction)	74.5 ↑ 1.6	83.2 ↑ 0.9
+LayerScale, Stochastic Depth	75.4 ↑ 0.9	82.0 ↓ 1.2
+128k prototypes	76.6 ↑ 1.2	81.9 ↓ 0.1
+KoLeo	78.9 ↑ 2.3	82.5 ↑ 0.6
+SwiGLU FFN	78.7 ↓ 0.2	83.1 ↑ 0.6
+Patch size 14	78.9 ↑ 0.2	83.5 ↑ 0.4
+Teacher momentum 0.994	79.4 ↑ 0.5	83.6 ↑ 0.1
+Tweak warmup schedules	80.5 ↑ 1.1	83.8 ↑ 0.2
+Batch size 3k	81.7 ↑ 1.2	84.7 ↑ 0.9
+Sinkhorn-Knopp	81.7 =	84.7 =
+Untying heads = DINOv2	82.0 ↑ 0.3	84.5 ↓ 0.2

Table 1: **Ablation study of the training differences between iBOT and DINOv2.** We optimize for k-NN performance, as in our experience, the linear probe performance is lower-bounded by the k-NN performance. Some modifications, like LayerScale and a high Stochastic Depth (rate=0.4), incur a decrease in linear probe performance, but have the benefits of increasing the stability of training by avoiding NaN loss values during training (Touvron et al., 2022). Overall, these modifications allowed for the next set of improvements to be added. Experiments are run using the ViT-Large architecture on ImageNet-22k.

Training Data	INet-1k	Im-A	ADE-20k	Oxford-M	iNat2018	iNat2021	Places205
INet-22k	**85.9**	73.5	46.6	62.5	81.1	85.6	67.0
INet-22k \ INet-1k	85.3	70.3	46.2	58.7	80.1	85.1	66.5
Uncurated data	83.3	59.4	48.5	54.3	68.0	76.4	67.2
LVD-142M	85.8	**73.9**	**47.7**	**64.6**	**82.3**	**86.4**	**67.6**

Table 2: **Ablation of the source of pretraining data.** We compare the INet-22k dataset that was used in iBOT to our dataset, LVD-142M. Each model is trained for the same number of iterations, that is smaller than in our final run, without high-resolution adaptation. Pretraining on LVD-142M maintains the performance over INet-1k while leading to models that perform better in other domains.

and a linear probe in Table 1. Generally, we observe that each component improves the performance on either k-NN or linear probing and even both in most cases. Only LayerScale and Stochastic Depth incur a performance drop in linear probing but significantly improve the training stability in our experience.

6.2 Pretraining Data Source

The quality of features is directly related to the quality of the pretraining data. In this experiment, we probe the impact of LVD-142M compared to ImageNet-22k, a commonly used pretraining dataset, or using directly raw and uncurated data. For the uncurated dataset, we randomly sample 142 million images from the same data source as LVD-142M. We train a ViT-g/14 on each dataset for the same number of iterations. We also include a variant of ImageNet-22k obtained by removing the synsets of ImageNet-1k (INet-22k \ INet-1k) for completeness. We report the comparisons in Table 2.

The most salient observation is that training on a curated set of images works better on most benchmarks than training on uncurated data. This confirms the benefit of curating data, even in the case of self-supervised pretraining. When compared with models trained on ImageNet-22k, training on LVD-142M is also superior on all the benchmarks but ImageNet-1k. This confirms that training on a more diverse set of images improves the quality of the features in domains that are not covered by ImageNet-22k. We also see that training on our curated data increases the performances on domains that are not used for the curation process (INaturalist 2018, 2021 and Places205), proving that scale and diversity can benefit unseen domains.

	INet-1k k-NN	INet-1k linear
iBOT	72.9	82.3
+(our reproduction)	74.5 ↑ 1.6	83.2 ↑ 0.9
+LayerScale, Stochastic Depth	75.4 ↑ 0.9	82.0 ↓ 1.2
+128k prototypes	76.6 ↑ 1.2	81.9 ↓ 0.1
+KoLeo	78.9 ↑ 2.3	82.5 ↑ 0.6
+SwiGLU FFN	78.7 ↓ 0.2	83.1 ↑ 0.6
+Patch size 14	78.9 ↑ 0.2	83.5 ↑ 0.4
+Teacher momentum 0.994	79.4 ↑ 0.5	83.6 ↑ 0.1
+Tweak warmup schedules	80.5 ↑ 1.1	83.8 ↑ 0.2
+Batch size 3k	81.7 ↑ 1.2	84.7 ↑ 0.9
+Sinkhorn-Knopp	81.7 =	84.7 =
+Untying heads = DINOv2	82.0 ↑ 0.3	84.5 ↓ 0.2

表1：iBOT与DINOv2训练差异的消融研究。我们以k-NN性能为优化目标，因为根据我们的经验，线性探测性能的下限由k-NN性能决定。某些修改（如LayerScale和高随机深度率=0.4）会导致线性探测性能下降，但能通过避免训练期间出现NaN损失值来提高训练稳定性（Touvron等人，2022）。总体而言，这些修改为后续改进奠定了基础。实验采用ViT-Large架构在ImageNet-22k数据集上进行。

Training Data	INet-1k	Im-A	ADE-20k	Oxford-M	iNat2018	iNat2021	Places205
INet-22k	**85.9**	73.5	46.6	62.5	81.1	85.6	67.0
INet-22k \ INet-1k	85.3	70.3	46.2	58.7	80.1	85.1	66.5
Uncurated data	83.3	59.4	48.5	54.3	68.0	76.4	67.2
LVD-142M	85.8	**73.9**	**47.7**	**64.6**	**82.3**	**86.4**	**67.6**

表2：预训练数据来源的消融研究。我们将iBOT中使用的INet-22k数据集与我们的LVD-142M数据集进行比较。每个模型均以相同迭代次数进行训练（此迭代次数少于最终实验），且未进行高分辨率适配。使用LVD-142M进行预训练能在保持INet-1k性能的同时，使模型在其他领域表现更优。

以及表1中的线性探测。总体而言，我们观察到每个组件都能提升k-NN或线性探测的性能，在大多数情况下甚至能同时提升两者。只有LayerScale和随机深度会导致线性探测性能下降，但根据我们的经验，它们显著提高了训练稳定性。

6.2 预训练数据来源

特征的质量与预训练数据的质量直接相关。在本实验中，我们探究了LVD-142M与常用预训练数据集ImageNet-22k或直接使用原始未筛选数据的影响对比。对于未筛选数据集，我们从与LVD-142M相同的数据源中随机抽取1.42亿张图像。我们在每个数据集上以相同的迭代次数训练ViT-g/14模型。为保持完整性，我们还纳入了通过移除ImageNet-1k同义词集（INet-22k {v*} INet-1k）得到的ImageNet-22k变体。对比结果如表2所示。

最显著的观察结果是，在精选图像集上进行训练在大多数基准测试中表现优于未筛选数据训练。这证实了数据筛选的益处，即使在自监督预训练的情况下也是如此。与在ImageNet-22k上训练的模型相比，使用LVD-142M进行训练在所有基准测试（除ImageNet-1k外）均表现更优。这表明在更多样化的图像集上训练能提升ImageNet-22k未覆盖领域的特征质量。我们还发现，使用精选数据训练能提升未参与筛选过程的领域性能（如INaturalist 2018、2021和Places205），证明规模与多样性对未见领域同样具有增益效果。

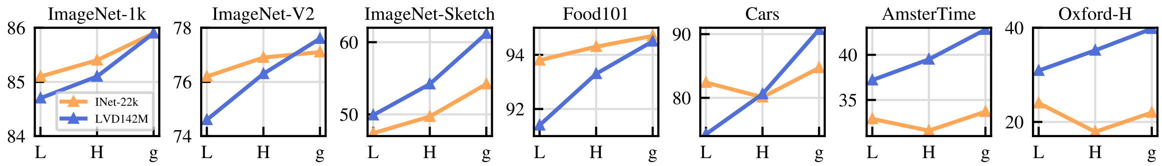


Figure 4: **Model scale versus data scale.** Evolution of performance as a function of model size for two different pretraining datasets: ImageNet-22k (14M images) and LVD-142M (142M images). The ViT-g trained on LVD-142M surpasses the ViT-g trained on ImageNet-22k on most benchmarks.

KoLeo	INet-1k	Im-A	ADE-20k	Oxford-M
×	85.3	70.6	47.2	55.6
✓	85.8	72.8	47.1	63.9

(a) Koleo loss

MIM	INet-1k	Im-A	ADE-20k	Oxford-M
×	85.3	72.0	44.2	64.3
✓	85.8	72.8	47.1	63.9

(b) MIM objective in iBOT

Table 3: **(a)** Effect of the KoLeo loss term. **(b)** Effect of the iBOT Masked Image Modeling (MIM) loss term. Evaluation performed on ImageNet-{1k,A} (classification with linear probe, accuracy %), ADE-20k (segmentation with linear layer, mIoU) and Oxford-M (image retrieval, mAP). Each model is trained on the same number of iterations, that is smaller than our final run. The KoLeo loss term improves nearest-neighbor search tasks (e.g. retrieval), and the MIM loss improves patch-level tasks (e.g. segmentation).

Overall, the conclusion of this ablation is that our dataset provides a good balance of different types of images that leads to the best performance overall.

6.3 Model Size and Data

We quantify the importance of scaling data with the model size in Fig. 4. As the size of models grow, training on LVD-142M becomes more beneficial than training on ImageNet-22k. For instance, a ViT-g trained on LVD-142M matches the performance on ImageNet-1k of a model trained on ImageNet-22k while significantly outperforming it on the other benchmarks.

6.4 Loss Components

We validated the proposed technical improvements in Sec. 6.1 by adding them incrementally. This section analyzes the performance hit observed if we ablate specific loss terms, starting from our best-performing model. We ablate the importance of the KoLeo loss and the impact of the masked image modeling term. For both, we report performance on ImageNet-1k using a linear classifier, ADE-20k segmentation using a linear classifier, and nearest-neighbor image retrieval on Oxford-M. Table 3a shows the impact of using the KoLeo loss. We see that the instance retrieval performance improves by more than 8%, confirming that this term helps spread features in the output space. At the same time, the other metrics do not suffer from this regularization. In Table 3b, we show the impact of using the masked image modeling term from iBOT. This term is critical for dense prediction tasks, leading to almost 3% performance improvement.

6.5 Impact of Knowledge Distillation

For small architectures, we distill larger models instead of training them from scratch. We use the distillation procedure described in Sec. 5. We evaluate the effectiveness of this approach by comparing a ViT-L/14 trained from scratch with one distilled from a ViT-g/14 over 12 benchmarks in Fig. 5. We also report the performance of the ViT-g/14 used for distillation as a topline. The distilled model outperforms the one trained from scratch on all 12 benchmarks, validating our pretraining approach for small models.

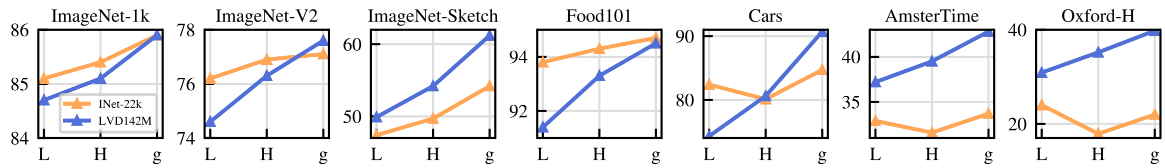


图4：模型规模与数据规模的关系。展示了两种不同预训练数据集（ImageNet-22k（1400万张图像）和LVD-142M（1.42亿张图像））下，模型大小对性能的影响。在大多数基准测试中，基于LVD-142M训练的ViT-g超越了基于ImageNet-22k训练的ViT-g。

KoLeo	INet-1k	Im-A	ADE-20k	Oxford-M
×	85.3	70.6	47.2	55.6
✓	85.8	72.8	47.1	63.9

(a) Koleo loss

MIM	INet-1k	Im-A	ADE-20k	Oxford-M
×	85.3	72.0	44.2	64.3
✓	85.8	72.8	47.1	63.9

(b) MIM objective in iBOT

表3：(a) KoLeo损失项的效果。(b) iBOT掩码图像建模(MIM)损失项的效果。评估在ImageNet-{1k,A}（线性探针分类，准确率%）、ADE-20k（线性层分割，mIoU）和Oxford-M（图像检索，mAP）上进行。每个模型均以相同迭代次数训练，该次数少于我们的最终运行。KoLeo损失项提升了最近邻搜索任务（例如检索）的性能，而MIM损失则改善了块级任务（例如分割）的表现。

总的来说，本次消融实验的结论是，我们的数据集在各类图像之间提供了良好的平衡，从而实现了整体最佳性能。

6.3 模型规模与数据

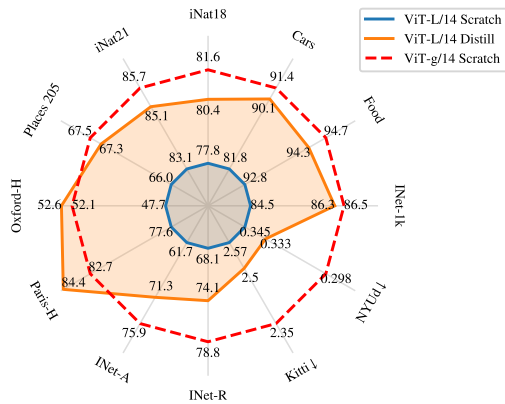
我们在图4中量化了数据规模随模型大小扩展的重要性。随着模型规模的增长，在LVD-142M上训练比在ImageNet-22k上训练更具优势。例如，在LVD-142M上训练的ViT-g模型在ImageNet-1k上的表现与在ImageNet-22k上训练的模型相当，同时在其他基准测试中显著优于后者。

6.4 损失分量

我们在第6.1节中通过逐步添加所提出的技术改进进行了验证。本节从我们性能最佳的模型出发，分析了若去除特定损失项所观察到的性能影响。我们分别去除了KoLeo损失的重要性以及掩码图像建模项的影响。针对这两者，我们报告了使用线性分类器在ImageNet-1k上的性能、使用线性分类器在ADE-20k分割任务上的表现，以及在Oxford-M数据集上的最近邻图像检索结果。表3a展示了使用KoLeo损失的影响。我们看到实例检索性能提升了超过8%，证实了该项损失有助于在输出空间中分散特征。同时，其他指标并未受到此正则化的负面影响。表3b展示了采用iBOT中的掩码图像建模项的影响。该项对密集预测任务至关重要，带来了近3%的性能提升。

6.5 知识蒸馏的影响

对于小型架构，我们采用蒸馏大型模型而非从头训练的方法。我们使用第5节中描述的蒸馏流程。通过在图5中比较从头训练的ViT-L/14与从ViT-g/14蒸馏得到的模型在12个基准测试上的表现，我们评估了该方法的有效性。同时，我们报告了用于蒸馏的ViT-g/14的性能作为上限参考。蒸馏模型在所有12个基准测试上均优于从头训练的模型，这验证了我们针对小型模型的预训练方法。



(a) Comparison on individual metrics

Arch	Method	INet-1k	Segm.	Depth↓	Classif.
ViT-g/14	Scratch	86.5	73.4	1.00	92.1
ViT-L/14	Scratch	84.5	72.2	1.10	90.2
ViT-L/14	Distill	**86.3**	**73.3**	**1.08**	**91.2**

Arch	Method	Finegr.	Retriev.	ARSketch	Video
ViT-g/14	Scratch	78.3	75.2	77.0	69.3
ViT-L/14	Scratch	75.8	71.3	69.5	67.3
ViT-L/14	Distill	**77.6**	**76.3**	**74.5**	**67.5**

(b) Averaged metrics on 8 vision tasks

Figure 5: **Effectiveness of knowledge distillation.** Comparison between a ViT-L trained from scratch or distilled from DINOv2 using ViT-g/14. For reference, we also report the performance of the ViT-g/14 teacher. We show that a ViT-L model distilled from a frozen ViT-g outperforms a the same model trained from scratch on all benchmarks, sometimes even outperforming the distillation target.

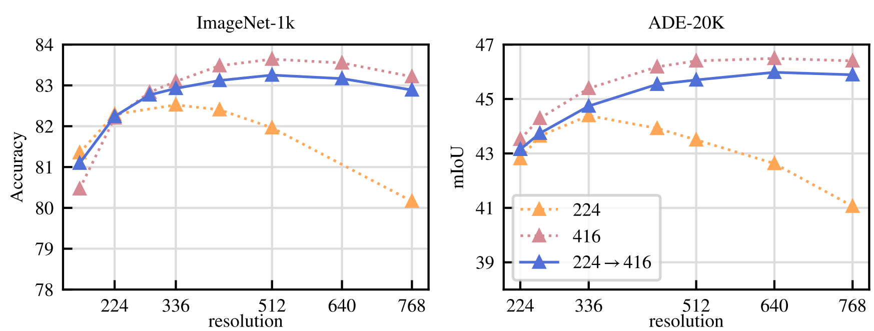
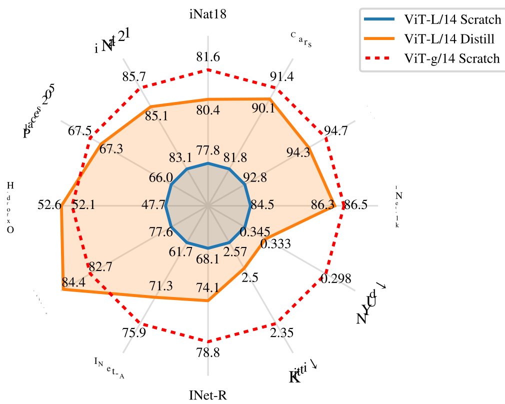


Figure 6: **Role of resolution.** Performance of ViT-L/16 trained on ImageNet-1k at fixed resolution ("224" and "416") or trained at 224 then 416 for a short duration ("224→416"). We train linear classifiers on top of frozen features at different resolutions and report Top-1 accuracy on ImageNet and mIoU on ADE-20k. We observe that performing SSL training at high resolution for a short duration achieve behavior and results close to training at the same high resolution for the full training, at a fraction of the cost.

6.6 Impact of Resolution

We measure the impact of changing the resolution during the pretraining on the performance of image and patch-level features. We consider models trained from scratch using a fixed resolution of either 224×224 or 416×416, and a model trained from scratch at 224×224, then resumed for 10k more iterations at 416×416. High-resolution training is compute-intensive, so we conduct this ablation on a small setup: a ViT-L/16 trained on ImageNet1k. In Fig. 6, we report the performance of a linear probe on ImageNet-1k and ADE-20k, evaluated at various resolutions. The model trained on high-resolution images performs best across resolutions, but this comes at a high cost: training at 416 is approximately $3\times$ more compute-intensive than training at 224. On the other hand, training at high resolution for only 10k iterations at the end of the training is almost as good and only requiring a fraction of the compute. As a consequence, we include this step at the end of the training rather than training at a high resolution from scratch.



(a) Comparison on individual metrics

Arch	Method	INet-1k	Segm.	Depth↓	Classif.
ViT-g/14	Scratch	86.5	73.4	1.00	92.1
ViT-L/14	Scratch	84.5	72.2	1.10	90.2
ViT-L/14	Distill	**86.3**	**73.3**	**1.08**	**91.2**

Arch	Method	Finegr.	Retriev.	ARSketch	Video
ViT-g/14	Scratch	78.3	75.2	77.0	69.3
ViT-L/14	Scratch	75.8	71.3	69.5	67.3
ViT-L/14	Distill	**77.6**	**76.3**	**74.5**	**67.5**

(b) Averaged metrics on 8 vision tasks

图5：知识蒸馏的有效性。比较了从头开始训练的ViT-L与使用ViT-g/14从DINOv2蒸馏得到的ViT-L。作为参考，我们还报告了ViT-g/14教师模型的性能。实验表明，通过冻结的ViT-g蒸馏得到的ViT-L模型在所有基准测试中均优于从头训练的相同模型，有时甚至超越了蒸馏目标本身。

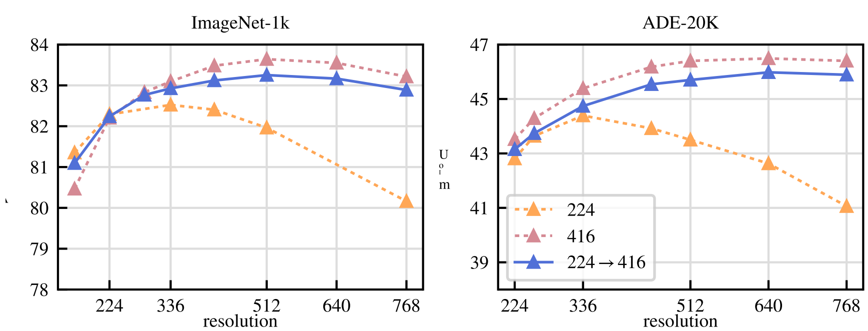


图6：分辨率的作用。在固定分辨率（“224”和“416”）下训练的ViT-L/16，或在224分辨率训练后短时间在416分辨率训练（“224→416”）的性能。我们在不同分辨率的冻结特征上训练线性分类器，并报告ImageNet上的Top-1准确率和ADE-20k上的mIoU。我们观察到，以高分辨率进行短时间的SSL训练，能以极低的成本获得与全程在同一高分辨率下训练相近的行为和结果。

6.6 分辨率的影响

我们测量了在预训练期间改变分辨率对图像和补丁级特征性能的影响。我们考虑了使用固定分辨率224×224或416×416从头开始训练的模型，以及先在224×224分辨率下从头训练，随后在416×416分辨率下继续训练1万次迭代的模型。高分辨率训练计算密集，因此我们在小规模设置下进行此消融实验：使用ViT-L/16架构在ImageNet1k上进行训练。在图6中，我们报告了在ImageNet-1k和ADE-20k上以不同分辨率评估的线性探测性能。使用高分辨率图像训练的模型在所有分辨率下表现最佳，但代价高昂：416分辨率训练的计算强度约为224分辨率训练的3倍。另一方面，仅在训练末期进行1万次迭代的高分辨率训练效果接近，且仅需少量额外计算。因此，我们在训练末期加入这一步骤，而非从头开始进行高分辨率训练。

7 Results

In this section, we present the empirical evaluation of our models on many image understanding tasks. We evaluate both global and local image representations, on category and instance-level recognition, semantic segmentation, monocular depth prediction, and action recognition. We detail the list of benchmarks in Appendix C. The goal of this evaluation is twofold. First, we show that our self-supervised features outperform the current state of the art by a very large margin. Second, we show that they match, or surpass the performance of weakly-supervised ones on a substantial number of tasks.

Baselines. In our comparisons, we use two kinds of models as baselines. We compare to the best performing self-supervised models that are openly available. First, we run our evaluations for MAE (He et al., 2022), DINO (Caron et al., 2021), SEERv2 (Goyal et al., 2022a), MSN (Assran et al., 2022), EsViT (Li et al., 2022a), Mugs (Zhou et al., 2022b) and iBOT (Zhou et al., 2022a). When several architectural variants were proposed for a given method, we report results for the one that leads to best top-1 accuracy on ImageNet-1k. Second, we report performance of open-source weakly-supervised models such as CLIP (Radford et al., 2021), OpenCLIP (Ilharco et al., 2021; Cherti et al., 2023), and SWAG (Singh et al., 2022). When evaluating models on ImageNet-1k, we report the performance for each of the aforementioned methods. For all other evaluations, we report the four best-performing models amongst SSL ones. Also, for reference, we report the best performing OpenCLIP-G for weakly-supervised ones.

7.1 ImageNet Classification

As a first evaluation, we probe the quality of the holistic image representation produced by the model on the ImageNet-1k classification dataset. We evaluate the quality of features by training a simple classifier over a frozen backbone, and do not perform finetuning of the backbone weights. Following previous work, we use a linear model for simplicity, ensuring a reproducible evaluation, despite the fact that classes may not be linearly separable. Because most SSL methods were developped using ImageNet-1k validation performance as a debugging signal, we also report the top-1 accuracy on ImageNet-ReaL and ImageNet-V2. In order to report this additional validation performance, for all models, we run the evaluation with our code. We compare our frozen features to the best publicly available SSL features in Table 4, regardless of architecture or pretraining data. We see the components proposed in this work lead to a very significant improvement (+4.2%) over the previous state of the art (iBOT ViT-L/16 trained on ImageNet-22k) on linear evaluation. At the same time, we also see that the performance increase on the alternative test sets is larger for our method, indicating stronger generalization. We describe details of our linear evaluation in Appendix B.3.

How far are we from weakly-supervised models? We also want to validate that our features are competitive with state-of-the-art open-source weakly supervised models. To this end, we compare on ImageNet-1k, using the linear evaluation, to three off-the-shelf methods with several architectural variants. For all models, we run the linear evaluation using our code, after making sure that our numbers match those reported in technical reports and papers. We show the result of this evaluation in Table 4. We see that our backbone, surpases the performance of OpenCLIP with a ViT-G/14 architecture (+0.3%) and EVA-CLIP with a ViT-g/14 (+0.1%). At the same time, we also observe that our performance on the ImageNet-V2 test set is significantly better (+1.1% versus EVA-CLIP), indicating better generalization. For the remainder of this section, we report OpenCLIP-G as a reference for weakly-supervised models.

Can we finetune the encoders? We question if the ability of our models to produce high quality frozen features impact their performance when finetuned with supervision on a specific dataset. While this is not core to this paper, this experiment is indicative of whether we have involuntarily specialized our models to the setting of linear evaluations of frozen features. To run this sanity check, we apply the finetuning pipeline from Touvron et al. (2022), without tweaking hyper-parameters. In Table 5, we show that the Top-1 accuracy on the validation set of ImageNet-1k improves by more than +2% when the backbone is finetuned. This is true both when using models at resolution 224 and 448. Further gains can be obtained by tuning the hyper-parameters of the finetuning, but this is beyond the goal of this sanity check. Nonetheless, our best finetuned performance (88.9%) is only a couple of percent below (−2.2%) the absolute state of the

7 结果

在本节中，我们展示了模型在多项图像理解任务上的实证评估结果。我们分别在类别与实例级识别、语义分割、单目深度预测以及动作识别任务中，评估了全局与局部图像表征的性能。具体基准测试列表详见附录C。本次评估的目标有两点：首先，我们证明自监督特征以显著优势超越当前最优方法；其次，我们表明这些特征在大量任务上达到甚至超越了弱监督方法的性能。

基准模型。在我们的比较中，我们使用两类模型作为基准。我们与公开可用的性能最佳的自监督模型进行比较。首先，我们对MAE（He等人，2022）、DINO（Caron等人，2021）、SEER.（Goyal等人，2022a）、MSN（Assran等人，2022）、EsViT（Li等人，2022a）、Mugs（Zhou等人，2022b）和iBOT（Zhou等人，2022a）进行评估。当某个方法提出多种架构变体时，我们报告在ImageNet-1k上获得最佳top-1准确率的变体结果。其次，我们报告开源弱监督模型的性能，例如CLIP（Radford等人，2021）、OpenCLIP（Ilharco等人，2021；Cherti等人，2023）和SWAG（Singh等人，2022）。在ImageNet-1k上评估模型时，我们报告上述每种方法的性能。对于所有其他评估，我们报告自监督模型中表现最佳的四个模型。此外，作为参考，我们报告弱监督模型中性能最佳的OpenCLIP-G。

7.1 ImageNet 图像分类

作为初步评估，我们在ImageNet-1k分类数据集上检验模型生成的整体图像表征质量。我们通过基于冻结主干网络训练简单分类器来评估特征质量，且不进行主干网络权重的微调。遵循先前工作，为简化流程并确保评估可复现，我们采用线性模型进行评估，尽管类别可能并非线性可分。由于大多数自监督学习方法在开发过程中使用ImageNet-1k验证性能作为调试信号，我们还报告了在ImageNet-ReaL和ImageNet-V2数据集上的top-1准确率。为获取这些额外的验证性能数据，我们对所有模型使用统一代码进行评估。如表4所示，我们将冻结特征与当前公开的最佳自监督学习特征进行比较（不考虑架构或预训练数据差异）。本工作提出的组件在线性评估中相较先前最优方法（基于ImageNet-22k训练的iBOT ViT-L/16）实现了显著提升（+4.2%）。同时，我们的方法在替代测试集上表现出更大的性能提升，表明其具有更强的泛化能力。线性评估的详细设置见附录B.3。

我们距离弱监督模型还有多远？我们同样希望验证我们的特征与当前最先进的开源弱监督模型相比具有竞争力。为此，我们在ImageNet-1k上使用线性评估方法，与三种现成方法及其多种架构变体进行了比较。对于所有模型，我们均使用自己的代码进行线性评估，并确保所得数据与技术报告和论文中报告的结果一致。我们在表4中展示了此次评估的结果。可以看到，我们的骨干网络性能超越了采用ViT-G/14架构的OpenCLIP（+0.3%）以及采用ViT-g/14的EVA-CLIP（+0.1%）。同时，我们还观察到在ImageNet-V2测试集上我们的表现明显更优（相比EVA-CLIP提升+1.1%），这表明了更好的泛化能力。在本节后续部分，我们将以OpenCLIP-G作为弱监督模型的参考基准。

我们能否对编码器进行微调？我们质疑模型生成高质量冻结特征的能力是否会影响其在特定数据集上进行有监督微调时的性能。虽然这不是本文的核心内容，但该实验能表明我们是否无意中将模型专门优化用于冻结特征的线性评估场景。为了进行这项验证，我们采用了Touvron等人（2022）的微调流程，未调整超参数。表5显示，当骨干网络被微调后，ImageNet-1k验证集的Top-1准确率提升了超过+2%。这一结论在224和448分辨率模型中都成立。通过调整微调超参数可获得更大增益，但这已超出本次验证的目标。尽管如此，我们最佳的微调性能（88.9%）仅比绝对最优状态低几个百分点（–2.2%）。

Method	Arch.	Data	Text sup.	kNN val	linear val	linear ReaL	linear V2
Weakly supervised							
CLIP	ViT-L/14	WIT-400M	✓	79.8	84.3	88.1	75.3
CLIP	ViT-L/14_{336}	WIT-400M	✓	80.5	85.3	88.8	75.8
SWAG	ViT-H/14	IG3.6B	✓	82.6	85.7	88.7	77.6
OpenCLIP	ViT-H/14	LAION-2B	✓	81.7	84.4	88.4	75.5
OpenCLIP	ViT-G/14	LAION-2B	✓	83.2	86.2	89.4	77.2
EVA-CLIP	ViT-g/14	custom*	✓	**83.5**	86.4	89.3	77.4
Self-supervised							
MAE	ViT-H/14	INet-1k	×	49.4	76.6	83.3	64.8
DINO	ViT-S/8	INet-1k	×	78.6	79.2	85.5	68.2
SEERv2	RG10B	IG2B	×	–	79.8	–	–
MSN	ViT-L/7	INet-1k	×	79.2	80.7	86.0	69.7
EsViT	Swin-B/W=14	INet-1k	×	79.4	81.3	87.0	70.4
Mugs	ViT-L/16	INet-1k	×	80.2	82.1	86.9	70.8
iBOT	ViT-L/16	INet-22k	×	72.9	82.3	87.5	72.4
DINOv2	ViT-S/14	LVD-142M	×	79.0	81.1	86.6	70.9
	ViT-B/14	LVD-142M	×	82.1	84.5	88.3	75.1
	ViT-L/14	LVD-142M	×	**83.5**	86.3	89.5	78.0
	ViT-g/14	LVD-142M	×	**83.5**	**86.5**	**89.6**	**78.4**

Table 4: **Linear evaluation on ImageNet-1k of frozen pretrained features.** We report Top-1 accuracy on the validation set for publicly available models trained on public or private data, and with or without text supervision (text sup.). For reference, we also report the kNN performance on the validation set. We compare across any possible architectures (Arch.), at resolution 224 × 224 unless stated otherwise. The dataset used for training EVA-CLIP is a custom mixture, see paper for details (Fang et al., 2023).

arts (91.1%), obtained by Chen et al. (2023a). As DINOv2 leads to features that are strong in both the linear and finetuning settings, a strong property of our approach is that *finetuning is optional*.

Arch.	Res.	Linear	Finetuned	Δ
ViT-g/14	224	86.5	88.5	+2.0
	448	86.7	88.9	+2.2

Table 5: **Supervised finetuning on ImageNet-1k.** We use the pipeline of Touvron et al. (2022) to finetune our encoders on ImageNet-1k at resolutions 224 × 224 or 448 × 448. We compare with the accuracy obtained with linear probing and observe only modest improvements with fine-tuning: this suggests that DINOv2 features already perform well out-of-the-box.

Robustness analysis. To complement our study, and probe the generalization of our features, we evaluate our ImageNet-1k models trained with linear classification heads on domain generalization benchmarks. We use the best performing linear classifier as described above and simply run inference on those benchmarks. Please note that most results in the literature are obtained with models that are finetuned end-to-end on ImageNet-1k. We show the result of this experiment in Table 6. When comparing with state-of-the-art SSL methods, our models shows drastically better robustness (+29.6% on A (Hendrycks et al., 2021b), +22.1% on R (Hendrycks et al., 2021a) and +23.0% on Sketch (Wang et al., 2019) compared to iBOT). Our model also improves upon the best weakly-supervised model on ImageNet-A while lagging behind on R and Sketch.

				kNN	linear		
Method	Arch.	Data	Text sup.	val	val	ReaL	V2
Weakly supervised							
CLIP	ViT-L/14	WIT-400M	✓	79.8	84.3	88.1	75.3
CLIP	ViT-L/14_{336}	WIT-400M	✓	80.5	85.3	88.8	75.8
SWAG	ViT-H/14	IG3.6B	✓	82.6	85.7	88.7	77.6
OpenCLIP	ViT-H/14	LAION-2B	✓	81.7	84.4	88.4	75.5
OpenCLIP	ViT-G/14	LAION-2B	✓	83.2	86.2	89.4	77.2
EVA-CLIP	ViT-g/14	custom*	✓	**83.5**	86.4	89.3	77.4
Self-supervised							
MAE	ViT-H/14	INet-1k	✕	49.4	76.6	83.3	64.8
DINO	ViT-S/8	INet-1k	✕	78.6	79.2	85.5	68.2
SEERv2	RG10B	IG2B	✕	–	79.8	–	–
MSN	ViT-L/7	INet-1k	✕	79.2	80.7	86.0	69.7
EsViT	Swin-B/W=14	INet-1k	✕	79.4	81.3	87.0	70.4
Mugs	ViT-L/16	INet-1k	✕	80.2	82.1	86.9	70.8
iBOT	ViT-L/16	INet-22k	✕	72.9	82.3	87.5	72.4
DINOv2	ViT-S/14	LVD-142M	✕	79.0	81.1	86.6	70.9
	ViT-B/14	LVD-142M	✕	82.1	84.5	88.3	75.1
	ViT-L/14	LVD-142M	✕	**83.5**	86.3	89.5	78.0
	ViT-g/14	LVD-142M	✕	**83.5**	**86.5**	**89.6**	**78.4**

表4：基于冻结预训练特征的ImageNet-1k线性评估。我们报告了在公开或私有数据上训练、使用或不使用文本监督（text sup.）的公开可用模型在验证集上的Top-1准确率。作为参考，我们也报告了验证集上的kNN性能。我们比较了所有可能的架构（Arch.），除特别说明外分辨率均为224 × 224。训练EVA-CLIP所用的数据集为定制混合数据集，详见论文（Fang et al., 2023）。

艺术（91.1%），由Chen等人（2023a）获得。由于DINOv2在线性和微调设置下都能产生强大的特征，我们方法的一个突出优势在于*finetuning is optional*。

Arch.	Res.	Linear	Finetuned	Δ
ViT-g/14	224	86.5	88.5	+2.0
	448	86.7	88.9	+2.2

表5：ImageNet-1k上的监督微调。我们采用Touvron等人（2022）的流程，在224×224或448×448分辨率下对ImageNet-1k进行编码器微调。通过与线性探测获得的准确率对比，我们观察到微调仅带来有限提升：这表明DINOv2特征在未经调整时已具备良好性能。

鲁棒性分析。为了补充我们的研究并探究我们特征的泛化能力，我们在领域泛化基准上评估了使用线性分类头训练的ImageNet-1k模型。我们采用上述性能最佳的线性分类器，并直接在这些基准上进行推理。请注意，文献中的大多数结果是通过在ImageNet-1k上端到端微调的模型获得的。我们在表6中展示了该实验的结果。与最先进的SSL方法相比，我们的模型展现出显著更强的鲁棒性（在A基准（Hendrycks等人，2021b）上提升+29.6%，在R基准（Hendrycks等人，2021a）上提升+22.1%，在Sketch基准（Wang等人，2019）上提升+23.0%，均优于iBOT）。我们的模型在ImageNet-A基准上超越了最佳弱监督模型，但在R和Sketch基准上仍略有落后。

Method	Arch	Data	Im-A	Im-R	Im-C↓	Sketch
OpenCLIP	ViT-G/14	LAION-2B	63.8	**87.8**	45.3	**66.4**
MAE	ViT-H/14	INet-1k	10.2	34.4	61.4	21.9
DINO	ViT-B/8	INet-1k	23.9	37.0	56.6	25.5
iBOT	ViT-L/16	INet-22k	41.5	51.0	43.9	38.5
DINOv2	ViT-S/14	LVD-142M	33.5	53.7	54.4	41.2
	ViT-B/14	LVD-142M	55.1	63.3	42.7	50.6
	ViT-L/14	LVD-142M	71.3	74.4	31.5	59.3
	ViT-g/14	LVD-142M	**75.9**	78.8	**28.2**	62.5

Table 6: **Domain Generalization with a linear probe** on top of frozen features at a resolution of 224. Higher numbers are better for all benchmarks except Im-C.

		Image classification			Video classification		
Feature	Arch	iNat2018	iNat2021	Places205	K400	UCF-101	SSv2
OpenCLIP	ViT-G/14	73.0	76.0	**69.8**	78.3	90.7	35.8
MAE	ViT-H/14	31.0	32.3	52.4	54.2	70.6	29.2
DINO	ViT-B/8	59.6	68.3	60.4	64.5	85.0	32.6
iBOT	ViT-L/16	66.3	74.6	64.4	72.6	88.6	**38.7**
DINOv2	ViT-S/14	69.0	74.2	62.9	67.8	87.0	33.1
	ViT-B/14	76.4	81.1	66.2	73.2	89.1	34.4
	ViT-L/14	80.4	85.1	67.3	76.3	90.5	35.6
	ViT-g/14	**81.6**	**85.7**	67.5	**78.4**	**91.2**	38.3

Table 7: **Linear evaluation on other image and video classification.** The image benchmarks contain a large quantity of fine-grained examples about objects or scenes. The video benchmarks cover action classification and human-object interaction. All the features are frozen with a linear probe on top.

7.2 Additional Image and Video classification Benchmarks

In this section we study the generalization of our features on downstream classification benchmarks. We consider two sets of evaluations in that context. On one hand, we use large and finegrained datasets such as iNaturalist and Places205. On the other, we use the 12 image classification tasks originally proposed in SimCLR (Chen et al., 2020). For iNaturalist 2018, iNaturalist 2021, and Places205, we train a linear classifier with data augmentations as in Sec. 7.1 We report top-1 accuracy for those three datasets in Table 7. Interestingly, our model significantly outperforms OpenCLIP ViT-G/14 on both variants of iNaturalist (+8.6% and +9.7% for 2018 and 2021 respectively), and lags slightly behind on Places 205 (−2.3%).

In a second set of evaluations, we measure the performance of our model on video action recognition even though our features were not trained on videos.. We evaluated features on three datasets, namely UCF-101 (Soomro et al., 2012), Kinetics-400 (Kay et al., 2017) and Something-Something v2 (Goyal et al., 2017). For this evaluation, we pick 8 evenly spaced frames in the video and train a linear classifier on the average of the features for UCF and K-400. For SSv2, we opt for concatenation to retain more temporal information than with feature averaging. For each dataset, we measure average accuracy and report the results in Table 7. We see that amongst self-supervised approaches, our model clearly sets a new state of the art. Moreover, our model matches the accuracy of the OpenCLIP features on UCF and Kinetics (+0.1% and +0.5% respectively) and clearly outperforms them on SSv2 (+2.5%). This is particularly interesting, as SSv2 requires a much richer understanding of the video frames.

Finally, in Table 8, we compare selected frozen features on 12 transfer classification benchmarks initially proposed by Chen et al. (2020). This benchmark covers scenes, objects (food, cars, planes), and textures.

Method	Arch	Data	Im-A	Im-R	Im-C↓	Sketch
OpenCLIP	ViT-G/14	LAION-2B	63.8	**87.8**	45.3	**66.4**
MAE	ViT-H/14	INet-1k	10.2	34.4	61.4	21.9
DINO	ViT-B/8	INet-1k	23.9	37.0	56.6	25.5
iBOT	ViT-L/16	INet-22k	41.5	51.0	43.9	38.5
DINOv2	ViT-S/14	LVD-142M	33.5	53.7	54.4	41.2
	ViT-B/14	LVD-142M	55.1	63.3	42.7	50.6
	ViT-L/14	LVD-142M	71.3	74.4	31.5	59.3
	ViT-g/14	LVD-142M	**75.9**	78.8	**28.2**	62.5

表6：在分辨率为224的冻结特征上使用线性探针的领域泛化结果。除Im-C外，所有基准测试中数值越高越好。

		Image classification			Video classification		
Feature	Arch	iNat2018	iNat2021	Places205	K400	UCF-101	SSv2
OpenCLIP	ViT-G/14	73.0	76.0	**69.8**	78.3	90.7	35.8
MAE	ViT-H/14	31.0	32.3	52.4	54.2	70.6	29.2
DINO	ViT-B/8	59.6	68.3	60.4	64.5	85.0	32.6
iBOT	ViT-L/16	66.3	74.6	64.4	72.6	88.6	**38.7**
DINOv2	ViT-S/14	69.0	74.2	62.9	67.8	87.0	33.1
	ViT-B/14	76.4	81.1	66.2	73.2	89.1	34.4
	ViT-L/14	80.4	85.1	67.3	76.3	90.5	35.6
	ViT-g/14	**81.6**	**85.7**	67.5	**78.4**	**91.2**	38.3

表7：其他图像和视频分类的线性评估。图像基准包含大量关于物体或场景的细粒度样本。视频基准涵盖动作分类及人物-物体交互。所有特征均被冻结，仅在其上使用线性探针。

7.2 额外的图像与视频分类基准测试

在本节中，我们研究了所提特征在下游分类基准上的泛化能力。我们在此背景下考虑了两组评估。一方面，我们使用了如iNaturalist和Places205这样大规模且细粒度的数据集；另一方面，我们采用了SimCLR（Chen等人，2020）最初提出的12项图像分类任务。对于iNaturalist 2018、iNaturalist 2021和Places205，我们按照第7.1节的方法使用数据增强训练线性分类器，并在表7中报告了这三个数据集的top-1准确率。有趣的是，我们的模型在两个版本的iNaturalist上均显著优于OpenCLIP ViT-G/14（2018年和2021年分别达到+8.6%和+9.7%），而在Places205上略逊一筹（−2.3%）。

在第二组评估中，我们测量了模型在视频动作识别上的性能，尽管我们的特征并非基于视频训练。我们在三个数据集上评估了特征，即UCF-101（Soomro等人，2012）、Kinetics-400（Kay等人，2017）和Something-Something v2（Goyal等人，2017）。在此评估中，我们从视频中选取8个均匀间隔的帧，并对UCF和K-400的特征平均值训练线性分类器。对于SSv2，我们选择特征拼接而非平均，以保留更多时序信息。针对每个数据集，我们计算平均准确率并在表7中报告结果。我们发现，在自监督方法中，我们的模型明显创造了新的最优性能。此外，我们的模型在UCF和Kinetics上的准确率与OpenCLIP特征持平（分别为+0.1%和+0.5%），并在SSv2上显著超越它们（+2.5%）。这一点尤其值得关注，因为SSv2需要对视频帧有更丰富的理解。

最后，在表8中，我们比较了Chen等人（2020）最初提出的12个迁移分类基准上选取的冻结特征。该基准涵盖场景、物体（食物、汽车、飞机）和纹理。

Feature	Arch	Food	C10	C100	SUN	Cars	Aircr	VOC	DTD	Pets	Cal101	Flowers	CUB	Avg
OpenCLIP	ViT-G/14	94.5	98.7	91.0	**84.0**	**96.1**	80.2	**89.3**	**86.0**	95.7	**98.1**	99.5	89.9	91.9
MAE	ViT-H/14	78.4	96.1	83.9	63.9	56.1	63.4	84.3	75.4	89.4	95.9	92.3	57.2	78.0
DINO	ViT-B/8	85.1	97.2	86.9	70.3	76.6	70.6	86.7	79.6	93.2	95.4	97.6	81.7	85.1
iBOT	ViT-L/16	91.0	99.0	92.8	75.6	71.8	72.4	89.0	80.7	87.7	97.5	99.6	82.1	86.6
DINOv2	ViT-S/14	89.1	97.7	87.5	74.4	81.6	74.0	87.8	80.6	95.1	97.0	99.6	88.1	87.7
	ViT-B/14	92.8	98.7	91.3	77.3	88.2	79.4	88.2	83.3	96.2	96.1	99.6	89.6	90.1
	ViT-L/14	94.3	99.3	93.4	78.7	90.1	81.5	88.3	84.0	96.6	97.5	99.7	90.5	91.2
	ViT-g/14	**94.7**	**99.5**	**94.4**	78.7	91.4	**87.2**	89.0	84.5	**96.7**	97.6	**99.7**	**91.6**	**92.1**

Table 8: **Linear evaluation of frozen features on fine-grained benchmarks.** Accuracy on 12 benchmarks covering objects, scenes and textures following the evaluation protocol proposed in Chen et al. (2020).

		Oxford		Paris		Met			AmsterTime
Feature	Arch	M	H	M	H	GAP	GAP-	ACC	mAP
OpenCLIP	ViT-G/14	50.7	19.7	79.2	60.2	6.5	23.9	34.4	24.6
MAE	ViT-H/14	11.7	2.2	19.9	4.7	7.5	23.5	30.5	4.2
DINO	ViT-B/8	40.1	13.7	65.3	35.3	17.1	37.7	43.9	24.6
iBOT	ViT-L/16	39.0	12.7	70.7	47.0	25.1	54.8	58.2	26.7
DINOv2	ViT-S/14	68.8	43.2	84.6	68.5	29.4	54.3	57.7	43.5
	ViT-B/14	72.9	49.5	90.3	78.6	36.7	63.5	66.1	45.6
	ViT-L/14	**75.1**	**54.0**	**92.7**	**83.5**	**40.0**	68.9	71.6	**50.0**
	ViT-g/14	73.6	52.3	92.1	82.6	36.8	**73.6**	**76.5**	46.7

Table 9: **Evaluation of frozen features on instance-level recognition.** We consider 4 different benchmarks and report their main metrics.

We replace the Birdsnap dataset with CUB because the former was not publicly available in its entirety. We follow the experimental protocol as outlined by Chen et al. (2020), namely training a logistic regression on precomputed features. Our model significantly outperforms state-of-the-art SSL models, with most notable differences on Stanford Cars (+14.8% versus DINO ViT-B/8) and FGVC Aircraft (+14.8% versus iBOT ViT-L/16). Even though these benchmarks favor text-guided pretraining, our features are still competitive with OpenCLIP on most classification benchmarks, with the exception of a few datasets, especially SUN (−5.3%) and Cars (−4.7%).

7.3 Instance Recognition

In this experiment, we probe our model on the task of instance-level recognition using a non-parametric approach. Images from a database are ranked according to their cosine similarity with a query image. We evaluated our model and compare to baselines on Paris and Oxford, that are landmark recognition benchmarks. We also evaluated on Met, a dataset of artworks from the Metropolitan museum, and AmsterTime, containing street view images matched to archival images of Amsterdam. We measure performance by computing the mean average precision and report our results in Table 9. We see that our features significantly outperform both SSL (+41% mAP on Oxford-Hard), and weakly-supervised (+34% mAP on Oxford-Hard) ones. It is interesting to see that our features perform well across task granularities, both at the category-level and instance-level. This is a desirable property for strong off-the-shelf computer vision features.

Feature	Arch	Food	C10	C100	SUN	Cars	Aircr	VOC	DTD	Pets	Cal101	Flowers	CUB	Avg
OpenCLIP	ViT-G/14	94.5	98.7	91.0	**84.0**	**96.1**	80.2	**89.3**	**86.0**	95.7	**98.1**	99.5	89.9	91.9
MAE	ViT-H/14	78.4	96.1	83.9	63.9	56.1	63.4	84.3	75.4	89.4	95.9	92.3	57.2	78.0
DINO	ViT-B/8	85.1	97.2	86.9	70.3	76.6	70.6	86.7	79.6	93.2	95.4	97.6	81.7	85.1
iBOT	ViT-L/16	91.0	99.0	92.8	75.6	71.8	72.4	89.0	80.7	87.7	97.5	99.6	82.1	86.6
DINOv2	ViT-S/14	89.1	97.7	87.5	74.4	81.6	74.0	87.8	80.6	95.1	97.0	99.6	88.1	87.7
	ViT-B/14	92.8	98.7	91.3	77.3	88.2	79.4	88.2	83.3	96.2	96.1	99.6	89.6	90.1
	ViT-L/14	94.3	99.3	93.4	78.7	90.1	81.5	88.3	84.0	96.6	97.5	99.7	90.5	91.2
	ViT-g/14	**94.7**	**99.5**	**94.4**	78.7	91.4	**87.2**	89.0	84.5	**96.7**	97.6	**99.7**	**91.6**	**92.1**

表8：在细粒度基准测试中对冻结特征的线性评估。遵循Chen等人（2020）提出的评估协议，在涵盖物体、场景和纹理的12个基准测试上的准确率。

		Oxford		Paris		Met			AmsterTime
Feature	Arch	M	H	M	H	GAP	GAP-	ACC	mAP
OpenCLIP	ViT-G/14	50.7	19.7	79.2	60.2	6.5	23.9	34.4	24.6
MAE	ViT-H/14	11.7	2.2	19.9	4.7	7.5	23.5	30.5	4.2
DINO	ViT-B/8	40.1	13.7	65.3	35.3	17.1	37.7	43.9	24.6
iBOT	ViT-L/16	39.0	12.7	70.7	47.0	25.1	54.8	58.2	26.7
DINOv2	ViT-S/14	68.8	43.2	84.6	68.5	29.4	54.3	57.7	43.5
	ViT-B/14	72.9	49.5	90.3	78.6	36.7	63.5	66.1	45.6
	ViT-L/14	**75.1**	**54.0**	**92.7**	**83.5**	**40.0**	68.9	71.6	**50.0**
	ViT-g/14	73.6	52.3	92.1	82.6	36.8	**73.6**	**76.5**	46.7

表9：基于冻结特征的实例级识别评估。我们考虑了4个不同的基准测试并报告其主要指标。

我们用CUB数据集替换了Birdsnap数据集，因为后者并未完全公开。我们遵循Chen等人（2020）提出的实验方案，即在预计算特征上训练逻辑回归模型。我们的模型显著优于当前最先进的SSL模型，其中在Stanford Cars（+14.8%对比DINO ViT-B/8）和FGVC Aircraft（+14.8%对比iBOT ViT-L/16）数据集上表现差异最为突出。尽管这些基准测试更倾向于文本引导的预训练方法，但我们的特征在大多数分类基准上仍与OpenCLIP具有竞争力，仅在少数数据集上稍逊一筹，特别是SUN（−5.3%）和Cars（−4.7%）数据集。

7.3 实例识别

在本实验中，我们采用非参数化方法，在实例级识别任务上测试我们的模型。通过计算查询图像与数据库图像的余弦相似度对图像进行排序。我们在巴黎和牛津这两个地标识别基准数据集上评估了模型性能，并与基线方法进行了对比。同时还在大都会艺术博物馆艺术品数据集Met，以及由街景图像与阿姆斯特丹历史档案图像匹配构成的AmsterTime数据集上进行了评估。我们通过计算平均精度均值来衡量性能，结果汇总于表9。实验表明，我们的特征显著优于自监督学习方法（牛津困难集上+41% mAP）和弱监督方法（牛津困难集上+34% mAP）。值得注意的是，我们的特征在不同任务粒度（类别级与实例级）上均表现优异，这为构建强大的即用型计算机视觉特征提供了理想特性。

		ADE20k (62.9)		CityScapes (86.9)		Pascal VOC (89.0)	
Method	Arch.	lin.	+ms	lin.	+ms	lin.	+ms
OpenCLIP	ViT-G/14	39.3	46.0	60.3	70.3	71.4	79.2
MAE	ViT-H/14	33.3	30.7	58.4	61.0	67.6	63.3
DINO	ViT-B/8	31.8	35.2	56.9	66.2	66.4	75.6
iBOT	ViT-L/16	44.6	47.5	64.8	74.5	82.3	84.3
DINOv2	ViT-S/14	44.3	47.2	66.6	77.1	81.1	82.6
	ViT-B/14	47.3	51.3	69.4	80.0	82.5	84.9
	ViT-L/14	47.7	**53.1**	70.3	80.9	82.1	86.0
	ViT-g/14	**49.0**	53.0	**71.3**	**81.0**	**83.0**	**86.2**

Table 10: **Semantic segmentation on ADE20K, CityScapes and Pascal VOC with frozen features** and a linear classifier (lin.) and with multiscale (+ms). The absolute state of the art – from Wang et al. (2022), Liu et al. (2021) and Chen et al. (2018) respectively – are mentioned at the top of the Table. For reference, using the Mask2Former pipeline (Steiner et al., 2021) with a ViT-Adapter (Chen et al., 2023b) on top of our frozen ViT-g/14 backbone gives 60.2 mIoU on ADE-20k.

7.4 Dense Recognition Tasks

We probe the quality of patch-level features extracted from our network on several dense downstream tasks. We consider semantic image segmentation and monocular depth estimation in several settings and we conduct evaluations on multiple datasets for each.

Semantic segmentation. For our semantic segmentation evaluation, we consider two different setups. **Linear**: a linear layer is trained to predict class logits from a patch tokens. It is used to produce a low-resolution logit map (eg 32x32 for a model with patch size 16), which is then upsampled to full resolution (512x512) to obtain a segmentation map. This procedure is extremely simple but cannot easily produce high-resolution segmentations. **+ms**: a boosted version of the linear setup. We concatenate the patch tokens of the 4 last layers, use a larger image resolution of 640, and use multiscale test-time augmentations to improve the predictions. We report the performance of our model variants as well as the baselines on three datasets under the two setups in Table 10.

Our models show very good performance on all datasets and for all setups. Interestingly, our evaluation using **+ms** is on par with fully finetuning MAE with an Upernet decoder (53.0 versus 53.6 mIoU). This is surprising because we use a significantly simpler predictor. Also, our best model, when evaluated using the boosted recipe, almost matches the state of the art on Pascal VOC (86.2 versus 89.0 mIoU).

Frozen backbone in a SOTA pipeline. In a final experiment, we freeze our backbone, and plug it into a ViT-Adapter Chen et al. (2023b) with a Mask2former head (Cheng et al., 2022). We tune the weights of the adapter and head, but keep the backbone frozen, meaning 66% of the weights are frozen. This allows for a lighter segmentation training than full end-to-end fine-tuning. With this setup, we reach 60.2 mIoU on ADE20k, close to the competitive state of the art, standing at 62.9 mIoU (Wang et al., 2022). Although our setup for this experiment doesn't makes use of the optimisations described in Sec. 5, the segmentation training in this experiment took 28 hours on 16 V100 GPUs.

Depth estimation. In this experiment, we evaluate our patch-level features on three monocular depth estimation benchmarks: NYUd, KITTI and zero-shot transfer from NYUd to SUN3d. We follow the evaluation protocol of Li et al. (2022b). We consider three different setups for this evaluation. **lin. 1**: we extract the last layer of the frozen transformer and concatenate the `[CLS]` token to each patch token. Then we bi-linearly upsample the tokens by a factor of 4 to increase the resolution. Finally we train a simple linear layer using a classification loss by dividing the depth prediction range in 256 uniformly distributed bins and

		ADE20k (62.9)		CityScapes (86.9)		Pascal VOC (89.0)	
Method	Arch.	lin.	+ms	lin.	+ms	lin.	+ms
OpenCLIP	ViT-G/14	39.3	46.0	60.3	70.3	71.4	79.2
MAE	ViT-H/14	33.3	30.7	58.4	61.0	67.6	63.3
DINO	ViT-B/8	31.8	35.2	56.9	66.2	66.4	75.6
iBOT	ViT-L/16	44.6	47.5	64.8	74.5	82.3	84.3
DINOv2	ViT-S/14	44.3	47.2	66.6	77.1	81.1	82.6
	ViT-B/14	47.3	51.3	69.4	80.0	82.5	84.9
	ViT-L/14	47.7	**53.1**	70.3	80.9	82.1	86.0
	ViT-g/14	**49.0**	53.0	**71.3**	**81.0**	**83.0**	**86.2**

表10：在ADE20K、CityScapes和Pascal VOC数据集上使用冻结特征与线性分类器（lin.）以及多尺度（+ms）的语义分割结果。表中顶部列出了各自领域的绝对最优方法——分别来自Wang等人（2022）、Liu等人（2021）和Chen等人（2018）。作为参考，在我们的冻结ViT-g/14骨干网络上采用Mask2Former流程（Steiner等人，2021）并叠加ViT-Adapter（Chen等人，2023b），在ADE-20k上获得了60.2 mIoU。

7.4 密集识别任务

我们在多个密集下游任务中探究了从我们网络中提取的补丁级特征的质量。我们在多种设置下考虑了语义图像分割和单目深度估计，并针对每个任务在多个数据集上进行了评估。

语义分割。在我们的语义分割评估中，我们考虑了两种不同的设置。线性：训练一个线性层从图像块标记中预测类别对数概率。它用于生成低分辨率对数概率图（例如，对于块大小为16的模型，生成32x32的图），然后上采样到全分辨率（512x512）以获得分割图。这个过程极其简单，但难以生成高分辨率分割。+ms：线性设置的增强版本。我们拼接最后4层的图像块标记，使用更大的图像分辨率640，并采用多尺度测试时间增强来改进预测。我们在表10中报告了两种设置下，我们的模型变体以及基线在三个数据集上的性能。

我们的模型在所有数据集和所有设置上都表现出非常优异的性能。有趣的是，使用+ms进行评估时，我们的结果与使用Upernet解码器对MAE进行完全微调的效果相当（53.0对比53.6 mIoU）。这令人惊讶，因为我们使用的预测器要简单得多。此外，我们最好的模型在使用增强方案进行评估时，几乎达到了Pascal VOC上的最先进水平（86.2对比89.0 mIoU）。

SOTA流程中的冻结骨干网络。在最后一个实验中，我们冻结了骨干网络，并将其接入ViT-Adapter（Chen等人，2023b）与Mask2former头部（Cheng等人，2022）。我们调整了适配器和头部的权重，但保持骨干网络冻结，这意味着66%的权重被固定。相比完整的端到端微调，这种设置能实现更轻量的分割训练。在此配置下，我们在ADE20k数据集上取得了60.2 mIoU，接近当前62.9 mIoU的竞争性技术水平（Wang等人，2022）。尽管本实验未采用第5节所述的优化策略，但分割训练仍在16块V100 GPU上耗时28小时完成。

深度估计。在本实验中，我们在三个单目深度估计基准上评估了我们的块级特征：NYUd、KITTI以及从NYUd到SUN3d的零样本迁移。我们遵循Li等人（2022b）的评估协议。为此评估，我们考虑了三种不同的设置。lin. 1：我们提取冻结transformer的最后一层，并将`[CLS]`标记与每个块标记拼接。然后，我们将标记双线性上采样4倍以提高分辨率。最后，我们通过将深度预测范围划分为256个均匀分布的区间并使用分类损失来训练一个简单的线性层。

Method	Arch.	NYUd (0.330) lin. 1	lin. 4	DPT	KITTI (2.10) lin. 1	lin. 4	DPT	NYUd → SUN RGB-D (0.421) lin. 1	lin. 4	DPT
OpenCLIP	ViT-G/14	0.541	0.510	0.414	3.57	3.21	2.56	0.537	0.476	0.408
MAE	ViT-H/14	0.517	0.483	0.415	3.66	3.26	2.59	0.545	0.523	0.506
DINO	ViT-B/8	0.555	0.539	0.492	3.81	3.56	2.74	0.553	0.541	0.520
iBOT	ViT-L/16	0.417	0.387	0.358	3.31	3.07	2.55	0.447	0.435	0.426
DINOv2	ViT-S/14	0.449	0.417	0.356	3.10	2.86	2.34	0.477	0.431	0.409
	ViT-B/14	0.399	0.362	0.317	2.90	2.59	2.23	0.448	0.400	0.377
	ViT-L/14	0.384	0.333	0.293	2.78	2.50	2.14	0.429	0.396	0.360
	ViT-g/14	**0.344**	**0.298**	**0.279**	**2.62**	**2.35**	**2.11**	**0.402**	**0.362**	**0.338**

Table 11: **Depth estimation with frozen features**. We report performance when training a linear classifier on top of one (lin. 1) or four (lin. 4) transformer layers, as well, as the DPT decoder (DPT) of Ranftl et al. (2021). We report the RMSE metric on the 3 datasets. Lower is better. For reference, we report state-of-the-art results taken from Li et al. (2022b) on each benchmark on top of the Table.

use a linear normalization following Bhat et al. (2021). **lin. 4**: we use the same protocol that we use with one layer, but concatenate the tokens from layers $l = \{3, 6, 9, 12\}$ for ViT-S/B, $l = \{5, 12, 18, 24\}$ for ViT-L, and $l = \{10, 20, 30, 40\}$ for ViT-g. **DPT**: we use the DPT decoder (Ranftl et al., 2021) on top of our frozen models and setup a regression task. We scale the size of the head following the dimension of the features for each architecture. We show results for all baselines, all datasets and all setups in Table 11.

From this table, we see that our features clearly surpass the best SSL and WSL features available. It is interesting to see that iBOT features extracted from a ViT-L outperform the ones of OpenCLIP with a ViT-G. This observation supports the intuition that caption-based feature learning fails to learn subtle patterns like this one. Also, our model, with the DPT decoder and frozen backbone, matches or exceeds the performance of the recent work of Li et al. (2022b). Finally, the out-of-domain generalization result on SUN-RGBd shows that our features allow very good transfer between domains. A depth prediction module trained on indoor scenes from NYUd generalizes pretty well to the outdoor examples of SUN-RGBd.

7.5 Qualitative Results

In this final section of the empirical evaluation of our features, we propose a few qualitative analyses.

Semantic Segmentation and Depth Estimation. We show some qualitative results for our dense prediction evaluations: segmentation on ADE20K in Fig. 7 and depth estimation on NYUd, KITTI and SUN RGB-D in Fig. 7. We compare DINOv2 with OpenCLIP with a linear classifier on each dataset. While not perfect, the linear segmentation model using our DINOv2 backbone produces good results and behaves much better than the OpenCLIP one under this evaluation setup. Indeed, the segmentation mask produced by OpenCLIP-G shows many artifacts and disconnected components. The qualitative results on depth estimation clearly illustrate the quantitative gap between OpenCLIP and DINOv2. These results highlight that our features, as well as the features extracted from OpenCLIP, are able to linearly separate complex information such as depth, even though neither was trained with this type of information. However, our features lead to a much smoother depth estimation, with less artifacts. Some objects such as the chair on the SUN RGB-D image are completely ignored by OpenCLIP and correctly positioned using our features.

Out-of-distribution generalization. We show a few examples of applying the depth prediction and segmentation linear classifiers to out-of-distribution examples in Fig. 8. The qualitative results support our claim that our features transfer between domains. The quality of the depth and segmentation predicted for pictures of animals, or paintings is very good, even though the domains are very different.

		NYUd (0.330)			KITTI (2.10)			NYUd → SUN RGB-D (0.421)		
Method	Arch.	lin. 1	lin. 4	DPT	lin. 1	lin. 4	DPT	lin. 1	lin. 4	DPT
OpenCLIP	ViT-G/14	0.541	0.510	0.414	3.57	3.21	2.56	0.537	0.476	0.408
MAE	ViT-H/14	0.517	0.483	0.415	3.66	3.26	2.59	0.545	0.523	0.506
DINO	ViT-B/8	0.555	0.539	0.492	3.81	3.56	2.74	0.553	0.541	0.520
iBOT	ViT-L/16	0.417	0.387	0.358	3.31	3.07	2.55	0.447	0.435	0.426
DINOv2	ViT-S/14	0.449	0.417	0.356	3.10	2.86	2.34	0.477	0.431	0.409
	ViT-B/14	0.399	0.362	0.317	2.90	2.59	2.23	0.448	0.400	0.377
	ViT-L/14	0.384	0.333	0.293	2.78	2.50	2.14	0.429	0.396	0.360
	ViT-g/14	**0.344**	**0.298**	**0.279**	**2.62**	**2.35**	**2.11**	**0.402**	**0.362**	**0.338**

表11：使用冻结特征的深度估计。我们报告了在一个（lin. 1）或四个（lin. 4）Transformer层之上训练线性分类器，以及Ranftl等人（2021）的DPT解码器（DPT）时的性能。我们在3个数据集上报告了RMSE指标，数值越低越好。作为参考，我们在表格顶部列出了每个基准测试中来自Li等人（2022b）的最新研究成果。

采用Bhat等人（2021）的线性归一化方法。第4行：我们使用与单层相同的处理流程，但对于ViT-S/B模型，将第$l = \{3,、6,、9,、12\}$层的特征标记进行拼接；对于ViT-L模型，拼接第$l = \{5,、12,、18,、24\}$层；对于ViT-g模型，则拼接第$l = \{10,、20,、30,、40\}$层。DPT部分：我们在冻结模型基础上采用DPT解码器（Ranftl等人，2021）并建立回归任务。根据各架构的特征维度调整头部尺寸。所有基线方法、数据集及配置的结果均展示在表11中。

从该表中可以看出，我们的特征明显超越了当前最优的自监督学习和弱监督学习特征。有趣的是，从ViT-L提取的iBOT特征甚至优于基于ViT-G的OpenCLIP特征。这一观察印证了我们的直觉：基于标题的特征学习方法难以捕捉此类细微模式。此外，采用DPT解码器并冻结主干网络的模型，在性能上达到甚至超越了Li等人（2022b）的最新研究成果。最后，在SUN-RGBd数据集上的跨领域泛化结果表明，我们的特征能够实现出色的跨域迁移效果——在NYUd室内场景训练得到的深度预测模块，对SUN-RGBd的室外样本同样表现出优秀的泛化能力。

7.5 定性结果

在实证评估我们特征的最后一节中，我们提出了一些定性分析。

语义分割与深度估计。我们展示了密集预测评估的一些定性结果：图7中的ADE20K分割以及图7中NYUd、KITTI和SUN RGB-D的深度估计。我们在每个数据集上将DINOv2与搭载线性分类器的OpenCLIP进行对比。尽管不够完美，但使用DINOv2骨干网络的线性分割模型产生了良好结果，在此评估设置下的表现远优于OpenCLIP模型。实际上，OpenCLIP-G生成的分割掩码存在大量伪影和断裂区域。深度估计的定性结果清晰展示了OpenCLIP与DINOv2之间的量化差距。这些结果表明，我们的特征以及从OpenCLIP提取的特征能够线性分离深度等复杂信息，尽管两者均未使用此类信息进行训练。然而，我们的特征能实现更平滑的深度估计，且伪影更少。在SUN RGB-D图像中，诸如椅子等物体被OpenCLIP完全忽略，而使用我们的特征则能准确定位。

分布外泛化。我们在图8中展示了将深度预测和分割线性分类器应用于分布外样本的几个示例。定性结果支持我们的特征能够在不同领域间迁移的结论。尽管动物图片或绘画作品的领域差异巨大，但预测出的深度和分割质量仍然非常出色。

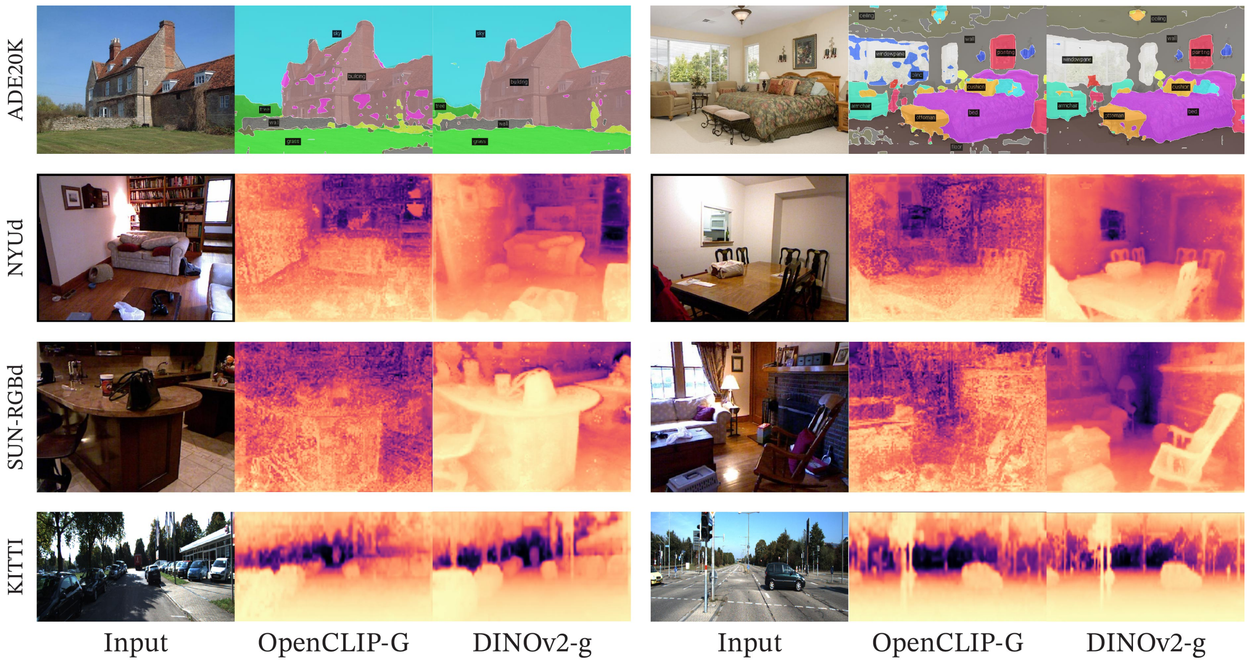


Figure 7: **Segmentation and depth estimation with linear classifiers.** Examples from ADE20K, NYUd, SUN RGB-D and KITTI with a linear probe on frozen OpenCLIP-G and DINOv2-g features.

Figure 8: **Examples of out-of-distribution examples** with frozen DINOv2-g features and a linear probe.

PCA of patch features. We show the results of the principal component analysis (PCA) performed on the patch features extracted by our model. We keep only patches with a positive value after we threshold the first component. This procedure turns out to separate the image's main object from the background. We compute a second PCA on the remaining patches across three images depicting the same category. We color the three first components with three different colors and present the results in Fig. 1 and 9. There are two interesting observations: first, our unsupervised foreground / background detector, based on detecting the highest variance direction, performs very well and is capable of delineating the boundary of the main object in the picture. Second, the other components correspond to "parts" of objects and match well for images of the same category. This is an emerging property – our model was not trained to parse parts of objects.

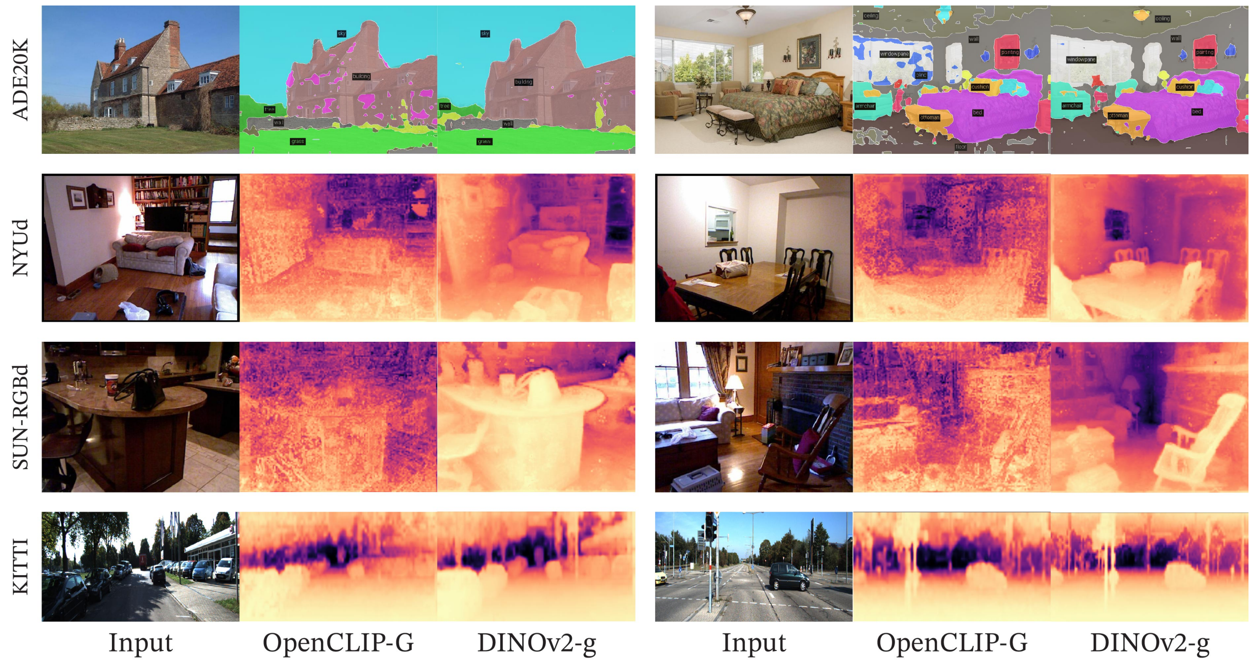


图7：使用线性分类器进行分割和深度估计。示例来自ADE20K、NYUd、SUN RGB-D和KITTI数据集，采用冻结的OpenCLIP-G和DINOv2-g特征进行线性探测。

图8：使用冻结的DINOv2-g特征和线性探针的分布外示例。

补丁特征的主成分分析。我们展示了基于模型提取的补丁特征进行主成分分析（PCA）的结果。我们仅保留第一主成分阈值化后具有正值的补丁。这一过程成功将图像主体与背景分离。随后，我们对三张同类图像中剩余的补丁进行第二次PCA分析。我们将前三个主成分分别用三种颜色标注，结果呈现在图1和图9中。有两个值得注意的发现：首先，我们基于最高方差方向检测的无监督前景/背景分离器表现优异，能够清晰勾勒出图像中主体的轮廓边界。其次，其他主成分对应物体的"部件"，在同类图像中具有良好的一致性。这是一种涌现特性——我们的模型并未经过物体部件解析的训练。

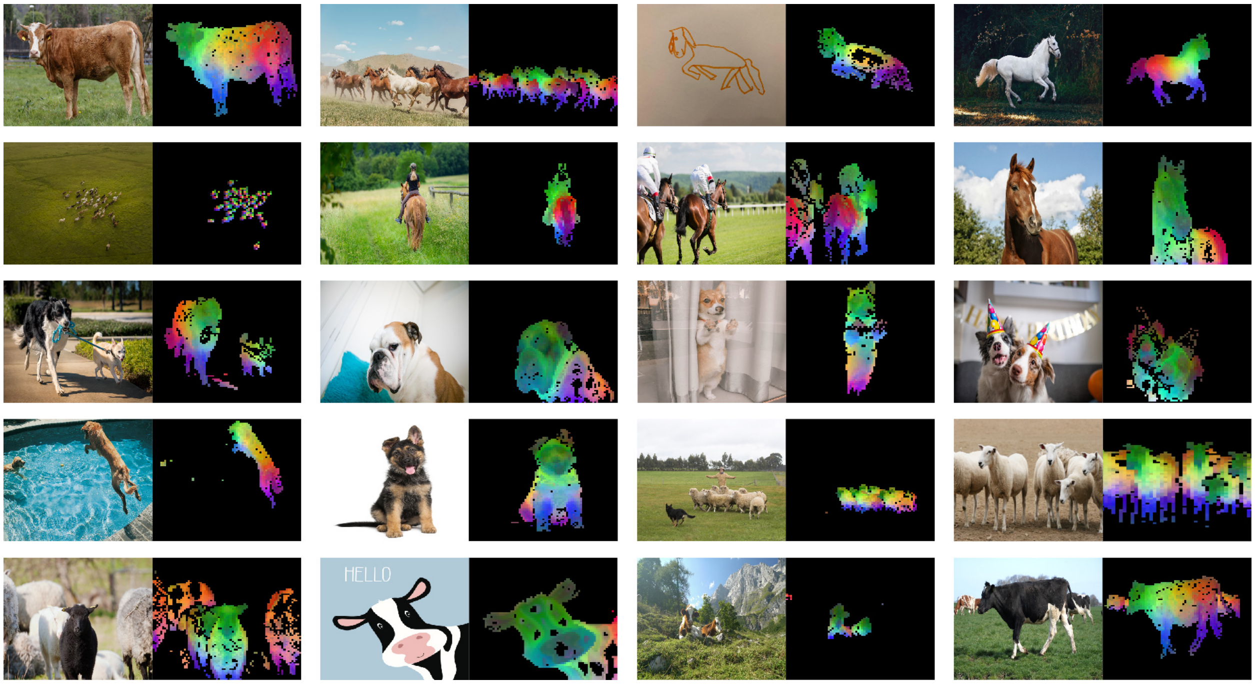


Figure 9: **More visualization of the first PCA components.** We compute the PCA between the patches from all of the images and show their first 3 components. Each component corresponds to a specific color channel. Same parts are matched between related images depsite changes of pose, style or even objects. Background is removed by removing patches with a negative score of the first PCA component.

Patch matching. Finally, we explore what type of information our patch-level features contain by matching them across images. We start by detecting the foreground object using the procedure described above. Then, we compute the euclidean distance between patch features extracted from two images and map them by solving an assignment problem. In order to reduce the number of matches, we then apply a non-maximum suppression to keep only the salient ones. In Fig. 10, we show some examples of such matchings.

We observe that the features seem to capture information about semantic regions that serve similar purpose in different objects or animals. For instance, the wing of a plane matches the wing of a bird. We also observe that the model is robust to style (image versus drawing), and to large variation of poses (see the elephant).

8 Fairness and Bias Analysis

We conduct two fairness evaluations of our models. We probe for geographical fairness and potential harmful label associations. For both evaluations, we experiment with our largest ViT-g model.

8.1 Geographical Fairness

We evaluate geographical fairness on the Dollar Street dataset introduced in De Vries et al. (2019) using the evaluation protocol of Goyal et al. (2022b). This benchmark compares performance across countries and income levels. It contains 16,073 images from 289 households across 54 countries. The task is to recognize 94 concepts that vary visually between households based on income or location. In Table 12, we compare our model with SEERv2 (Goyal et al., 2022a), a model trained on a geographically diverse set of images. Our model is slightly fairer across regions and incomes than the SEERv2 model and significantly better than the supervised baseline reported by Goyal et al. (2022a). However, we still observe a significant difference between regions, particularly in Africa, where our model performance drops by 25.7% compared to Europe. This shows that our model is still biased toward Western countries. Similarly, our model performs

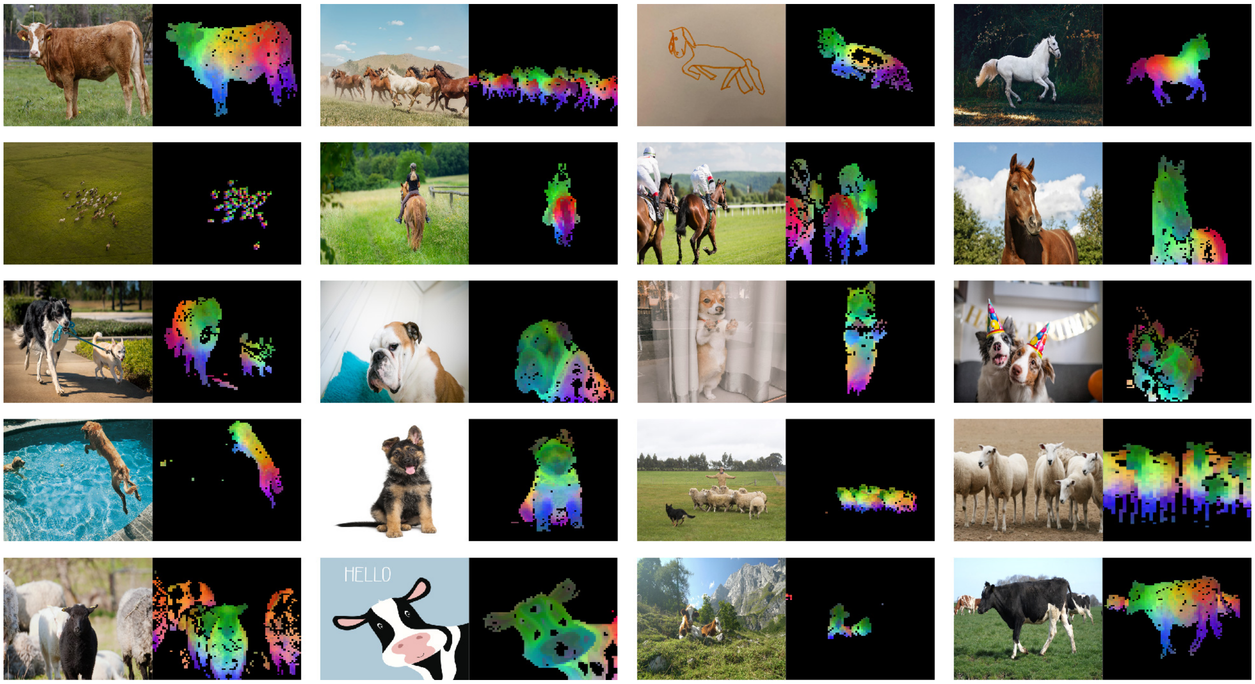


图9：第一主成分的更多可视化展示。我们计算了所有图像中图像块之间的主成分分析，并展示了它们的前三个成分。每个成分对应一个特定的颜色通道。尽管姿态、风格甚至物体发生变化，相关图像间的相同部分仍能匹配。通过移除第一主成分得分为负的图像块，背景已被去除。

补丁匹配。最后，我们通过跨图像匹配补丁级特征来探索它们包含何种信息。首先使用上述方法检测前景物体。接着，计算从两幅图像中提取的补丁特征之间的欧氏距离，并通过求解分配问题建立映射关系。为减少匹配数量，我们随后应用非极大值抑制仅保留显著匹配。图10展示了此类匹配的若干示例。

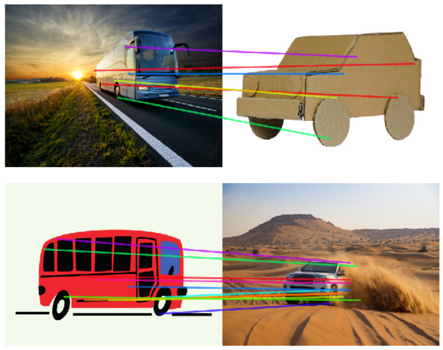
我们观察到，这些特征似乎捕捉到了在不同物体或动物中具有相似功能的语义区域信息。例如，飞机的机翼与鸟类的翅膀相匹配。我们还注意到，该模型对风格（图像与绘图）以及姿态的大幅变化（参见大象示例）具有鲁棒性。

8 公平性与偏见分析

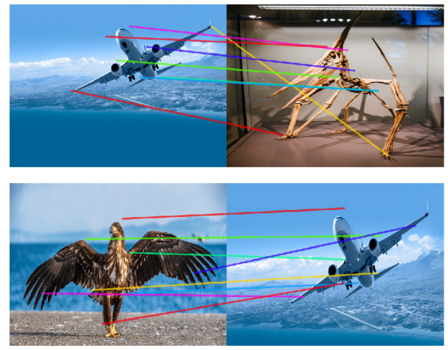
我们对模型进行了两项公平性评估。我们探究了地理公平性和潜在有害标签关联。在这两项评估中，我们均采用了最大的ViT-g模型进行实验。

8.1 地理公平性

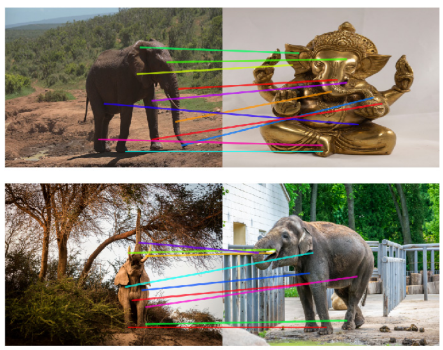
我们在De Vries等人（2019年）提出的Dollar Street数据集上，采用Goyal等人（2022b）的评估协议来评估地理公平性。该基准测试比较了不同国家和收入水平下的模型性能，包含来自54个国家289个家庭的16,073张图像，任务要求识别94个因家庭收入或地理位置而产生视觉差异的概念。在表12中，我们将模型与SEERv2（Goyal等人，2022a）进行对比——后者是基于地理多样性图像集训练的模型。我们的模型在跨地区和收入群体中表现出比SEERv2稍高的公平性，且显著优于Goyal等人（2022a）报告的有监督基线。然而，我们仍观察到地区间的显著差异，特别是在非洲地区，模型性能较欧洲下降了25.7%。这表明我们的模型仍存在对西方国家的偏向性。同样，我们的模型在……



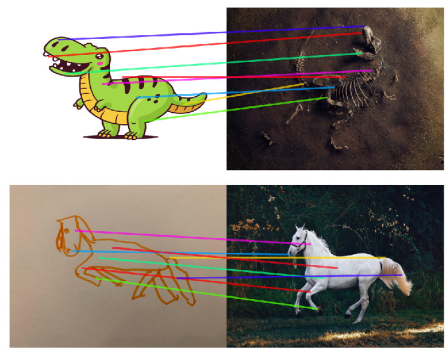
(Vehicles)



(Birds / Airplanes)



(Elephants)



(Drawings / Animals)

Figure 10: **Matching across images.** We match patch-level features between images from different domains, poses and even objects that share similar semantic information. This exhibits the ability of our model to transfer across domains and understand relations between similar parts of different objects.

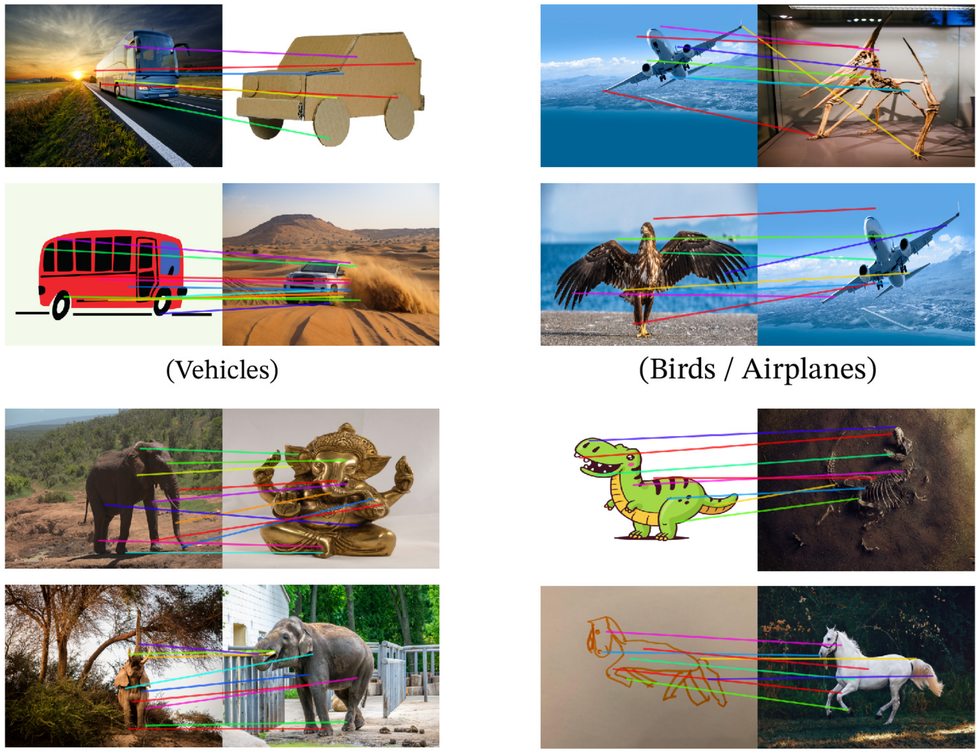
			Income buckets			**Regions**			
Method	Arch.	Data	low	medium	high	Africa	Asia	Americas	Europe
SEERv2	RG-10B	IG-1B	59.7	78.5	86.6	65.9	76.3	81.1	85.6
DINOv2	ViT-g/14	LVD-142M	67.4	83.3	90.5	74.0	81.6	86.2	89.7

Table 12: **Geographical fairness and diversity analysis across income buckets and regions.**

significantly better on high-income households than low-income ones, with a difference of 31.7%. Despite improvements, we observe significant biases in our models toward wealthy households from Western countries.

8.2 Gender, Skintones and Age

In a second set of evaluations, we question how our model classifies images of people of different gender, skin tone, and age (all self-reported). We follow the protocol of Goyal et al. (2022b), where we train a multiclass classifier on a subset of 619 classes of ImageNet-22k. We group the 619 classes into four broader categories: Human, Possibly Human, Non-Human, or Crime. Non-Human and Crime are considered harmful. Using this classifier, we run inference on 2955 images from the Casual Conversations dataset (Hazirbas et al., 2021) and keep all labels in the top-5 that are assigned a probability of 0.1 or more. Because of that, we can assign multiple classes to each image. We make one modification to the original evaluation protocol: we do not backpropagate gradients to the backbone and keep it frozen. We compare our model to SEERv2 in Table 13.



(Vehicles) (Birds / Airplanes)

(Elephants) (Drawings / Animals)

图10：跨图像匹配。我们在不同领域、姿态甚至共享相似语义信息的物体图像之间匹配块级特征。这展示了我们模型跨领域迁移以及理解不同物体相似部分之间关系的能力。

			Income buckets			Regions			
Method	Arch.	Data	low	medium	high	Africa	Asia	Americas	Europe
SEERv2	RG-10B	IG-1B	59.7	78.5	86.6	65.9	76.3	81.1	85.6
DINOv2	ViT-g/14	LVD-142M	67.4	83.3	90.5	74.0	81.6	86.2	89.7

表12：地理 跨收入群体的公平性和多样性分析 票券和区域。

对高收入家庭的改善效果明显优于低收入家庭，差异达31.7%。尽管有所改进，我们仍观察到模型明显偏向来自西方国家的富裕家庭。

8.2 性别、肤色与年龄

在第二组评估中，我们探讨了模型如何对不同性别、肤色和年龄（均为自我报告）的人物图像进行分类。我们遵循Goyal等人（2022b）的评估方案，在ImageNet-22k的619个类别子集上训练了一个多类别分类器，并将这些类别归纳为四大类：人类、可能人类、非人类或犯罪。其中非人类和犯罪类别被视为有害类别。使用该分类器，我们对Casual Conversations数据集（Hazirbas等人，2021）中的2955张图像进行推理，并保留前5个预测中概率达到0.1或以上的所有标签。因此，每张图像可能被赋予多个类别。我们对原始评估方案进行了一项调整：不向主干网络反向传播梯度，保持其冻结状态。我们在表13中将我们的模型与SEERv2进行了比较。

Model	Assoc.	Gender Skintone female darker	female lighter	male darker	male lighter	Age Groups 18-30	30-45	45-70	70+
SEER RG-10B	`Non-Human`	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	`Crime`	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	`Human`	94.9	95.8	86.6	79.0	90.5	88.3	91.9	82.3
	`Possibly-Human`	13.6	6.7	65.0	60.2	32.8	37.2	29.4	6.5
DINOv2 ViT-g/14	`Non-Human`	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	`Crime`	0.0	0.0	0.2	0.0	0.0	0.1	0.0	0.0
	`Human`	97.3	97.7	86.1	84.0	91.2	90.2	93.2	88.7
	`Possibly-Human`	15.8	17.2	52.2	48.1	35.3	37.3	23.0	9.7

Table 13: **Label association fairness evaluation across gender, skintones and age groups.** We follow the protocol proposed by Goyal et al. (2022b) with a slight modification. Instead of finetuning the backbone, we simply learn a linear classifier on the subset of 619 classes of ImageNet-22k.

Model to Reproduce	GPU Type	GPU Power consumption	GPU-hours	PUE	Total power consumption	Carbon emitted (tCO_2eq)
DINOv2-g	A100-40GB	400W	22,016	1.1	9.7 MWh	3.7

Table 14: **Carbon footprint of reproducing DINOv2.** We report the potential carbon emission of reproducing DINOv2-g when assuming a power consumption for the A100-40GB of 400W, a PUE of 1.1 and carbon intensity factor of 0.385 kg CO_2e per KWh.

Our model often classifies images of all groups as Human without large deviations across skin tones. Neither SEERv2 nor DINOv2 predict harmful labels from the Non-Human or Crime meta-categories (except for two instances where the background contains bars visually similar to prison bars). We see that our model triggers the Possibly-Human classes often. This class is constructed from objects in ImageNet-22k that are often related to Humans, such as Scarf, Glasses, or Beard. Our model often predicts the Possibly-Human class for men because of the prevalence of the Beard class. No clear pattern indicates a bias against a particular group in this study. While this is encouraging, we also acknowledge that a more thorough evaluation of biases may reveal flaws in our model.

9 Estimating the Environmental Impact of Training our Models

Training foundation models consumes a significant amount of energy, resulting in carbon dioxide emissions. Patterson et al. (2021) propose a methodology to report an estimation of the carbon emitted during the training of a model based on the specifics of the data center and its power grid. This computation informs the design of the data center used for the training of models and the choice of location for data centers. This methodology requires to know the specifics of the data center used for training, which can be complex when multiple data centers are involved over time. Additionally, these specifics are most often not in the control of the AI practitioner, and hence, this methodology is less helpful when practioners make technical decisions about future trainings. Instead, in this section, we follow an alternative that reports the potential carbon emission of retraining a similar model in an average data center located in the US. This methodology was used in previous work in natural language processing (Strubell et al., 2019; Touvron et al., 2023) to establish an apple-to-apple comparison between pretraining schemes. More precisely, we fix the value of all exogenous variables, i.e., the Power Usage Effectiveness (PUE) and carbon intensity factor of a power grid to the same values as in Touvron et al. (2023), that is, a PUE of 1.1 and the carbon intensity factor to the US average of 0.385 kg CO_2eq/KWh. We use the same formula as in Patterson et al. (2021) to estimate the potential energy consumption and the carbon emission. For the power consumption of an A100-80GB, we take the thermal design power for NVLink systems, which is 400W. We report the potential carbon emission

Model	Assoc.	Gender Skintone female darker	female lighter	male darker	male lighter	Age Groups 18-30	30-45	45-70	70+
SEER RG-10B	Non-Human	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Crime	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Human	94.9	95.8	86.6	79.0	90.5	88.3	91.9	82.3
	Possibly-Human	13.6	6.7	65.0	60.2	32.8	37.2	29.4	6.5
DINOv2 ViT-g/14	Non-Human	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	Crime	0.0	0.0	0.2	0.0	0.0	0.1	0.0	0.0
	Human	97.3	97.7	86.1	84.0	91.2	90.2	93.2	88.7
	Possibly-Human	15.8	17.2	52.2	48.1	35.3	37.3	23.0	9.7

表13：跨性别、肤色和年龄组的标签关联公平性评估。我们遵循Goyal等人（2022b）提出的方案，并稍作修改：不微调主干网络，仅在ImageNet-22k的619个子类上训练线性分类器。

Model to Reproduce	GPU Type	GPU Power consumption	GPU-hours	PUE	Total power consumption	Carbon emitted (tCO_2eq)
DINOv2-g	A100-40GB	400W	22,016	1.1	9.7 MWh	3.7

表14：DINOv2复现的碳足迹。我们假设A100-40GB功耗为400W、PUE为1.1、碳强度系数为每千瓦时0.385千克CO_2e，据此报告复现DINOv2-g的潜在碳排放量。

我们的模型通常将所有群体的图像分类为人类，且在不同肤色间没有显著偏差。SEERv2和DINOv2均未从“非人类”或“犯罪”元类别中预测出有害标签（除两例背景包含视觉上类似监狱栅栏的条状物外）。我们发现，我们的模型频繁触发“可能为人类”类别。该类别由ImageNet-22k中常与人类相关的物体（如围巾、眼镜或胡须）构成。由于“胡须”类别的高频出现，我们的模型常将男性图像预测为“可能为人类”类。本研究未发现明显针对特定群体的偏见模式。尽管这一结果令人鼓舞，我们也承认更深入的偏见评估可能揭示模型的缺陷。

9 评估训练我们模型的环境影响

训练基础模型会消耗大量能源，导致二氧化碳排放。Patterson等人（2021）提出了一种方法，根据数据中心及其电网的具体情况，报告模型训练过程中碳排放的估算值。这一计算为用于模型训练的数据中心设计以及数据中心选址提供了参考依据。该方法需要了解用于训练的数据中心的具体情况，当涉及多个数据中心且时间跨度较长时，情况可能变得复杂。此外，这些具体细节通常不受AI从业者控制，因此，当从业者对未来训练做出技术决策时，该方法的帮助有限。相反，在本节中，我们采用另一种方法，报告在美国一个平均数据中心重新训练类似模型的潜在碳排放量。该方法在自然语言处理领域的前期工作中已有应用（Strubell等人，2019；Touvron等人，2023），用于在不同预训练方案之间建立公平比较。具体而言，我们将所有外生变量（即电网的电力使用效率（PUE）和碳强度因子）固定为与Touvron等人（2023）相同的数值，即PUE为1.1，碳强度因子为美国平均值0.385千克CO_2eq/千瓦时。我们使用与Patterson等人（2021）相同的公式来估算潜在能耗和碳排放量。对于A100-80GB的功耗，我们采用NVLink系统的热设计功耗，即400W。我们报告潜在碳排放量

of retraining a DINOv2 ViT-g in Table 14. For comparison, retraining an OpenCLIP ViT-L or OpenCLIP ViT-G would require 22.4 MWh and 118.9 MWh, respectively, if run in the same data center. This is 10× more carbon emission. Note that this comparison is not fair to them, since they also train a text encoder in parallel, and we thus do not report them in the table. However, it gives a reasonable guideline for those who are interested in training only visual features: in this context, training a self-supervised model is preferable in terms of carbon emission. Training a text-guided model still makes sense when planning to reuse the text encoder.

Carbon footprint of the whole project. Additionally, we estimate the footprint of the whole project to be between 0.5k and 1k tCO_2eq using the same grid as presented above [3]. This carbon footprint represents in the order of 200k GPU-days. The primary sources of emissions are the self-supervised pre-trainings of the models. For example, a single pre-training of a ViT-g model (22k GPU-hours) emits 3.7 tons of CO_2eq, while a finetuning on ImageNet-1k (1k GPU-hours) emits 0.2 tons. This estimate only considers the GPUs' electricity consumption and ignores other emissions, such as their manufacturing and disposal.

10 Future work and Discussion

In this work, we present DINOv2, a new series of image encoders pretrained on large curated data with no supervision. This is the first SSL work on image data that leads to visual features that close the performance gap with (weakly) supervised alternatives across a wide range of benchmarks and without the need for finetuning. We can attribute the strong performance of the DINOv2 family of models to several factors: **i**) an improved training recipe with better hyperparameters and regularization (Table 1), **ii**) a larger model scale with improved results regardless of the data used for training (Fig. 4), **iii**) a larger dataset (Fig. 4) and **iv**) the distillation process that makes smaller models benefit from the performance of the strongest ViT-g model (Fig. 5). A few properties emerge from these models, such as an understanding of object parts and scene geometry regardless of the image domains. We expect that more of these properties will emerge at larger scales of models and data, akin to instruction emergence in large language models, and plan to continue scaling along these axes. This paper also demonstrates that these visual features are compatible with classifiers as simple as linear layers - meaning the underlying information is *readily available*. In future work, we plan to leverage this ability to train a a language-enabled AI system that can process visual features as if they were word tokens, and extract the required information to ground the system.

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[3]For context, a full Boeing 777 return flight between London and New York corresponds to approximately 560 tCO_2eq.

在表14中，我们展示了重新训练DINOv2 ViT-g的能耗。作为对比，若在同一数据中心运行，重新训练OpenCLIP ViT-L或OpenCLIP ViT-G将分别消耗22.4兆瓦时和118.9兆瓦时，这意味着会增加10×的碳排放量。需要注意的是，这种对比对他们而言并不公平，因为他们还同时训练了文本编码器，因此我们未在表中列出这些数据。然而，这为那些仅关注视觉特征训练的研究者提供了合理的参考：在此背景下，从碳排放角度考虑，训练自监督模型更为可取。当然，若计划复用文本编码器，训练文本引导模型仍然具有实际意义。

整个项目的碳足迹。此外，我们使用上述相同的电网估算整个项目的碳足迹在0.5k至1k吨CO_2当量之间[3]。这一碳足迹相当于约20万GPU日的能耗。排放的主要来源是模型的自监督预训练。例如，单次ViT-g模型的预训练（2.2万GPU小时）会排放3.7吨CO_2当量，而在ImageNet-1k上的微调（1千GPU小时）会排放0.2吨。此估算仅考虑了GPU的电力消耗，未包含其他排放源，例如设备制造与废弃处理。

10 未来工作与讨论

在本工作中，我们提出了DINOv2——一个基于大规模精选数据无监督预训练的新型图像编码器系列。这是首个在图像数据上实现自监督学习的研究，其视觉特征在广泛基准测试中弥合了与（弱）监督方法的性能差距，且无需微调。DINOv2模型家族的卓越性能可归因于以下因素：i) 改进的训练方案（包含更优超参数与正则化策略，见表1）；ii) 更大的模型规模（无论使用何种训练数据均能提升效果，见图4）；iii) 更庞大的数据集（图4）；iv) 蒸馏过程使较小模型能受益于最强ViT-g模型的性能（图5）。这些模型展现出若干特性，例如跨图像域理解物体部件与场景几何结构的能力。我们预期在更大规模的模型与数据中会涌现更多此类特性，类似于大语言模型中的指令涌现现象，并计划持续沿这些方向扩展规模。本文还证明这些视觉特征可与简单如线性层的分类器兼容——这意味着底层信息具有*readily available*特性。未来工作中，我们计划利用这种能力训练具备语言理解的AI系统，使其能像处理词元那样处理视觉特征，并从中提取必要信息以构建系统基础。

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A Data Processing

A.1 Data selection

Our selection of datasets for building LVD-142M is detailed in Tab. 15. This collection is intended to provide images covering well various downstream vision tasks both for image-level and dense recognition.

A.2 Image similarity

We employ cosine similarity to compare image features (whether ours or feature generated for deduplication) with the following similarity function m:

$$m(s, r) = \text{cosine-similarity}\left(f\left(s\right), f\left(r\right)\right) = \frac{f(s) \cdot f(r)}{\|f(s)\|_2 \|f(r)\|_2}$$

where s and r are a pair of images to compare and f is the model generating features.

A.3 Deduplication

Self-deduplication. To deduplicate our uncurated data source of 1.3B images, we compute and use the embeddings generated by Pizzi et al. (2022) and retrieve the $k = 64$ nearest neighbors of each image (using cosine similarity). Considering only neighbors with a similarity >0.6, we extract the connected components of the associated k-NN graph thanks to a scalable disjoint set data structure implementation. We then only keep one representative for each component of duplicate images. This results in a self-deduplicated data source of 1.1B images.

Relative deduplication To reduce redundancy and also properly evaluate the performance of our features, we discard remaining images of our self-deduplicated data source that are too similar to train and test splits of our evaluation datasets. To achieve this, we apply a similar procedure as for self-deduplication, with a stricter similarity >0.45, this time identifying the duplicate components (if any) to which each reference image belong and discarding it entirely. This results in a self- and relatively-deduplicated data source of 744M images.

A.4 Retrieval

We employ two approaches to augment dataset via retrieval: sample-based and cluster-based. The first one, sample-based, applies to datasets larger than 1M images and consists in collecting a fixed number k of nearest images for each sample image of the dataset to retrieve, effectively trying to multiply by k the size of the dataset. We use $k = 4$ for Google Landmarks v2 and ImageNet-22k but a larger $k = 32$ to make this specific retrieval a core part of our LVD-142M dataset. For smaller datasets, the second approach, cluster-based, consists in first clustering our uncurated data source into $100,000$ separate clusters thanks to a distributed k-means implementation. Each cluster should capture different types of image concept and contents. We then pick $10,000$ images from each cluster associated with more than 3 images of the retrieved dataset. As this can result in a very large number of retrieved images for some dataset, we restrict such retrievals to a maximum of 1M images to maintain the balance between the different datasets within LVD-142M.

B Implementation Details

B.1 Unsupervised pre-training

For unsupervised pre-training we build on the DINO and iBOT codebases. We use hyperparameters shown in Table 16, ViT architectures described in Table 17.

KoLeo regularization. We apply the KoLeo regularizer with a weight of 0.1 between the class tokens of the first global crop, for all samples within a GPU without cross-communication for this step.

数据处理

A.1 数据选择

我们为构建LVD-142M选择的数据集详情见表15。该数据集集合旨在提供全面覆盖图像级识别与密集识别下游视觉任务的多样化图像。

A.2 图像相似度

我们采用余弦相似度来比较图像特征（无论是我们的还是为去重生成的特征），使用以下相似度函数 m:

$$m(s,r) = \text{cosine-similarity}\left(f\left(s\right), f\left(r\right)\right) = \frac{f(s)\cdot f(r)}{\|f(s)\|_2\|f(r)\|_2}$$

其中 s 和 r 是一对需要比较的图像， f 是生成特征的模型。

A.3 去重

自去重。为了对我们未经筛选的13亿张图像数据源进行去重，我们计算并使用了Pizzi等人（2022年）生成的嵌入向量，检索每张图像的k =64个最近邻（使用余弦相似度）。仅考虑相似度>0.6的邻居，我们通过可扩展的并查集数据结构实现提取关联k-NN图的连通分量。随后，我们仅为每个重复图像分量保留一个代表图像。最终得到一个经过自去重处理的11亿张图像数据源。

相对去重 为了减少冗余并正确评估我们特征的性能，我们舍弃了自去重数据源中与评估数据集的训练和测试分割过于相似的剩余图像。为此，我们采用了与自去重类似的流程，但设置了更严格的相似度阈值>0.45，此次会识别每张参考图像所属的重复组件（如果存在）并将其完全剔除。最终我们得到了一个经过自去重和相对去重的数据源，包含7.44亿张图像。

A.4 检索

我们采用两种基于检索的数据增强方法：样本级和聚类级。第一种样本级方法适用于超过100万张图像的数据集，其核心是为待检索数据集中的每个样本图像收集固定数量k的最近邻图像，从而将数据集规模有效扩大k倍。对于Google Landmarks v2和ImageNet-22k数据集，我们设定k =为4；而在构建LVD-142M数据集时，则采用更大的k =值32，使该检索过程成为数据集的核心组成部分。对于较小规模的数据集，我们采用第二种聚类级方法：首先通过分布式k-均值算法将未筛选数据源划分为100,000个独立聚类，每个聚类应捕捉不同类型的图像概念与内容。随后从每个包含检索数据集3张以上图像的聚类中选取10,000张图像。考虑到某些数据集可能因此产生海量检索图像，我们将此类检索结果上限设定为100万张，以维持LVD-142M内部不同数据集间的平衡。

B 实现细节

B.1 无监督预训练

对于无监督预训练，我们基于DINO和iBOT的代码库进行构建。我们采用表16所示的超参数以及表17描述的ViT架构。

KoLeo正则化。我们在GPU内所有样本的第一个全局裁剪的类别标记之间应用权重为0.1的KoLeo正则化器，此步骤无需跨通信。

Task	Dataset / Split	Images	Retrieval	Retrieved	Final
classification	ImageNet-22k / –	14,197,086	as is	–	14,197,086
classification	ImageNet-22k / –	14,197,086	sample	56,788,344	56,788,344
classification	ImageNet-1k / train	1,281,167	sample	40,997,344	40,997,344
fine-grained classif.	Caltech 101 / train	3,030	cluster	2,630,000	1,000,000
fine-grained classif.	CUB-200-2011 / train	5,994	cluster	1,300,000	1,000,000
fine-grained classif.	DTD / train1	1,880	cluster	1,580,000	1,000,000
fine-grained classif.	FGVC-Aircraft / train	3,334	cluster	1,170,000	1,000,000
fine-grained classif.	Flowers-102 / train	1,020	cluster	1,060,000	1,000,000
fine-grained classif.	Food-101 / train	75,750	cluster	21,670,000	1,000,000
fine-grained classif.	Oxford-IIIT Pet / trainval	3,680	cluster	2,750,000	1,000,000
fine-grained classif.	Stanford Cars / train	8,144	cluster	7,220,000	1,000,000
fine-grained classif.	SUN397 / train1	19,850	cluster	18,950,000	1,000,000
fine-grained classif.	Pascal VOC 2007 / train	2,501	cluster	1,010,000	1,000,000
segmentation	ADE20K / train	20,210	cluster	20,720,000	1,000,000
segmentation	Cityscapes / train	2,975	cluster	1,390,000	1,000,000
segmentation	Pascal VOC 2012 (seg.) / trainaug	1,464	cluster	10,140,000	1,000,000
depth estimation	Mapillary SLS / train	1,434,262	as is	–	1,434,262
depth estimation	KITTI / train (Eigen)	23,158	cluster	3,700,000	1,000,000
depth estimation	NYU Depth V2 / train	24,231	cluster	10,850,000	1,000,000
depth estimation	SUN RGB-D / train	4,829	cluster	4,870,000	1,000,000
retrieval	Google Landmarks v2 / train (clean)	1,580,470	as is	–	1,580,470
retrieval	Google Landmarks v2 / train (clean)	1,580,470	sample	6,321,880	6,321,880
retrieval	AmsterTime / new	1,231	cluster	960,000	960,000
retrieval	AmsterTime / old	1,231	cluster	830,000	830,000
retrieval	Met / train	397,121	cluster	62,860,000	1,000,000
retrieval	Revisiting Oxford / base	4,993	cluster	3,680,000	1,000,000
retrieval	Revisiting Paris / base	6,322	cluster	3,660,000	1,000,000
					142,109,386

Table 15: **Composition of our LVD-142M dataset.** We report the list of datasets and associated splits used to build the dataset, how they were included (as is without retrieval or via sample-based or cluster-based retrieval). For retrievals, we indicate the actual number of retrieved images and the final number included in the dataset. We chose to include as many datasets as possible in the pretraining data in order to cover as many domains as possible. We kept a few datasets aside in order to evaluate performance outside of the pretraining domain. More details about dataset usages can be found in Table 18.

Task	Dataset / Split	Images	Retrieval	Retrieved	Final
classification	ImageNet-22k / –	14,197,086	as is	–	14,197,086
classification	ImageNet-22k / –	14,197,086	sample	56,788,344	56,788,344
classification	ImageNet-1k / train	1,281,167	sample	40,997,344	40,997,344
fine-grained classif.	Caltech 101 / train	3,030	cluster	2,630,000	1,000,000
fine-grained classif.	CUB-200-2011 / train	5,994	cluster	1,300,000	1,000,000
fine-grained classif.	DTD / train1	1,880	cluster	1,580,000	1,000,000
fine-grained classif.	FGVC-Aircraft / train	3,334	cluster	1,170,000	1,000,000
fine-grained classif.	Flowers-102 / train	1,020	cluster	1,060,000	1,000,000
fine-grained classif.	Food-101 / train	75,750	cluster	21,670,000	1,000,000
fine-grained classif.	Oxford-IIIT Pet / trainval	3,680	cluster	2,750,000	1,000,000
fine-grained classif.	Stanford Cars / train	8,144	cluster	7,220,000	1,000,000
fine-grained classif.	SUN397 / train1	19,850	cluster	18,950,000	1,000,000
fine-grained classif.	Pascal VOC 2007 / train	2,501	cluster	1,010,000	1,000,000
segmentation	ADE20K / train	20,210	cluster	20,720,000	1,000,000
segmentation	Cityscapes / train	2,975	cluster	1,390,000	1,000,000
segmentation	Pascal VOC 2012 (seg.) / trainaug	1,464	cluster	10,140,000	1,000,000
depth estimation	Mapillary SLS / train	1,434,262	as is	–	1,434,262
depth estimation	KITTI / train (Eigen)	23,158	cluster	3,700,000	1,000,000
depth estimation	NYU Depth V2 / train	24,231	cluster	10,850,000	1,000,000
depth estimation	SUN RGB-D / train	4,829	cluster	4,870,000	1,000,000
retrieval	Google Landmarks v2 / train (clean)	1,580,470	as is	–	1,580,470
retrieval	Google Landmarks v2 / train (clean)	1,580,470	sample	6,321,880	6,321,880
retrieval	AmsterTime / new	1,231	cluster	960,000	960,000
retrieval	AmsterTime / old	1,231	cluster	830,000	830,000
retrieval	Met / train	397,121	cluster	62,860,000	1,000,000
retrieval	Revisiting Oxford / base	4,993	cluster	3,680,000	1,000,000
retrieval	Revisiting Paris / base	6,322	cluster	3,660,000	1,000,000
					1.42亿

表15：我们的LVD-142M数据集构成。我们列出了用于构建该数据集的各数据集及其对应划分、纳入方式（直接采用未检索数据，或通过基于样本/聚类的检索纳入）。对于检索部分，我们标注了实际检索到的图像数量以及最终纳入数据集的图像数量。我们选择在预训练数据中纳入尽可能多的数据集，以覆盖尽可能多的领域。同时我们保留了部分数据集，用于评估预训练领域之外的性能表现。更多关于数据集使用的细节可在表18中找到。

	Arch.	Drop-rate	LR	Batch size
DINOv2-S (distilled)	ViT-S/14	0	1e-3	2048
DINOv2-B (distilled)	ViT-B/14	0	1e-3	2048
DINOv2-L (distilled)	ViT-L/14	0	1e-3	2048
DINOv2-L (from scratch)	ViT-L/14	0.4	3.5e-4	3072
DINOv2-g (from scratch)	ViT-g/14	0.4	3.5e-4	3072

Table 16: **Training hyperparameters for DINOv2-S, DINOv2-B, DINOv2-L and DINOv2-g.** All models run for 625k iterations with optimizer AdamW, an initial LayerScale value of 1e-5, a weight decay cosine schedule from 0.04 to 0.2, a learning rate warmup of 100k iterations, a teacher momentum cosine schedule from 0.994 to 1, and we train in float16 precision in all cases (except for the DINO heads where we reduce the gradients in float32).

Arch.	Embed dim	Heads	Blocks	FFN layer
ViT-S/14 (distilled)	384	6	12	MLP
ViT-B/14 (distilled)	768	12	18	MLP
ViT-L/14 (distilled)	1024	16	24	MLP
ViT-L/14 (from scratch)	1024	16	24	SwiGLU
ViT-g/14 (from scratch)	1536	24	40	SwiGLU

Table 17: **Architecture details of the ViT-S/B/L/g networks used in this work.** We use MLP feed-forward networks for distilled models, and SwiGLU (Shazeer, 2020) when training from scratch.

EMA update for the teacher. The teacher is initialized with the same state as the student, and is an exponential moving average of the student network, with a momentum value in [0.994, 1.0] following a cosine schedule. It is updated at the end of every training step.

B.2 High-Resolution adaptation

We initialise the model with the pretrained weights then train it for 10k iterations with the same procedure as the original pretraining. All the schedules are kept the same as in the original training, but compressed to fit in 10k iterations. All the hyperparameters are kept the same as in the first pretraining, except the base learning rate which is reduced.

B.3 Linear probing evaluation

For linear probing we define 3 evaluation parameters: the learning rate, how many output layers we use, whether we concatenate the average-pooled patch token features with the class token (or use only the class token). We train our linear layer with SGD for 12500 iterations, using random-resized-crop data augmentation, and perform the following grid search:

- learning rate in $\{0.0001, 0.0002, 0.0005, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.5\}$
- output layers in $\{1, 4\}$
- concatenate average-pooled tokens in $\{yes, no\}$

We then report the highest accuracy value obtained on the validation set as is common practice. Note that this grid search is not expensive, because at each iteration we perform inference on the backbone only once, then feed the output to all linear classifiers (each performing a single matrix multiplication).

C List of Datasets used

We show in Table 18 the list of benchmarks and datasets used and their purposes.

	Arch.	Drop-rate	LR	Batch size
DINOv2-S (distilled)	ViT-S/14	0	1e-3	2048
DINOv2-B (distilled)	ViT-B/14	0	1e-3	2048
DINOv2-L (distilled)	ViT-L/14	0	1e-3	2048
DINOv2-L (from scratch)	ViT-L/14	0.4	3.5e-4	3072
DINOv2-g (from scratch)	ViT-g/14	0.4	3.5e-4	3072

表16：DINOv2-S、DINOv2-B、DINOv2-L和DINOv2-g的训练超参数。所有模型均运行62.5万次迭代，使用AdamW优化器，初始LayerScale值为1e-5，权重衰减余弦调度从0.04到0.2，学习率预热10万次迭代，教师动量余弦调度从0.994到1，所有情况均采用float16精度训练（除DINO头部外，我们在该部分使用float32精度降低梯度）。

Arch.	Embed dim	Heads	Blocks	FFN layer
ViT-S/14 (distilled)	384	6	12	MLP
ViT-B/14 (distilled)	768	12	18	MLP
ViT-L/14 (distilled)	1024	16	24	MLP
ViT-L/14 (from scratch)	1024	16	24	SwiGLU
ViT-g/14 (from scratch)	1536	24	40	SwiGLU

表17：本工作中使用的ViT-S/B/L/g网络架构细节。对于蒸馏模型，我们使用MLP前馈网络；而在从头开始训练时，则采用SwiGLU（Shazeer, 2020）。

教师网络的EMA更新。教师网络以与学生网络相同的状态初始化，并且是学生网络的指数移动平均，动量值在[0.994, 1.0]之间，遵循余弦调度。它在每个训练步骤结束时进行更新。

B.2 高分辨率适配

我们使用预训练权重初始化模型，然后按照与原始预训练相同的流程进行10k次迭代训练。所有训练计划均保持与原始训练一致，但压缩至适配10k次迭代的规模。除基础学习率被降低外，所有超参数均与首次预训练保持相同。

B.3 线性探测评估

对于线性探测，我们定义了3个评估参数：学习率、使用的输出层数量，以及是否将平均池化的补丁令牌特征与类别令牌拼接（或仅使用类别令牌）。我们使用SGD训练线性层12500次迭代，采用随机缩放裁剪数据增强，并进行以下网格搜索：

- 学习率在 $\{0.0001, 0.0002, 0.0005, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.5\}$
- $\{1, 4\}$中的输出层
- 在$\{yes, no\}$中连接平均池化的标记

随后，我们按照惯例报告在验证集上获得的最高准确率值。需要注意的是，这种网格搜索并不耗费大量计算资源，因为在每次迭代中，我们仅对主干网络进行一次推理，随后将输出结果馈送至所有线性分类器（每个分类器仅执行一次矩阵乘法）。

C 使用的数据集列表

我们在表　　e 18 所使用的基准测试和数据集列表及其p　　目的。

Dataset	Pretraining (as is)	Retrieving pretraining data	Eval.	Task	Citation
ImageNet-1k	✗	✓	✓	Classif.	(Russakovsky et al., 2015)
ImageNet-22k	✓	✓	✗		(Deng et al., 2009)
ImageNet-V2	✗	✗	✓	Classif.	(Recht et al., 2019)
ImageNet-ReaL	✗	✗	✓	Classif.	(Beyer et al., 2020)
ImageNet-A	✗	✗	✓	Classif.	(Hendrycks et al., 2021b)
ImageNet-C	✗	✗	✓	Classif.	(Hendrycks & Dietterich, 2019)
ImageNet-R	✗	✗	✓	Classif.	(Hendrycks et al., 2021a)
ImageNet-Sk.	✗	✗	✓	Classif.	(Wang et al., 2019)
Food-101	✗	✓	✓	Classif.	(Bossard et al., 2014)
CIFAR-10	✗	✓	✓	Classif.	(Krizhevsky et al., 2009)
CIFAR-100	✗	✓	✓	Classif.	(Krizhevsky et al., 2009)
SUN397	✗	✓	✓	Classif.	(Xiao et al., 2010)
StanfordCars	✗	✓	✓	Classif.	(Krause et al., 2013)
FGVC-Aircraft	✗	✓	✓	Classif.	(Maji et al., 2013)
VOC 2007	✗	✓	✓	Classif.	(Everingham et al., 2010)
DTD	✗	✓	✓	Classif.	(Cimpoi et al., 2014)
Oxford Pets	✗	✓	✓	Classif.	(Parkhi et al., 2012)
Caltech101	✗	✓	✓	Classif.	(Fei-Fei et al., 2004)
Flowers	✗	✓	✓	Classif.	(Nilsback & Zisserman, 2008)
CUB200	✗	✓	✓	Classif.	(Welinder et al., 2010)
iNaturalist 2018	✗	✗	✓	Classif.	(Van Horn et al., 2018)
iNaturalist 2021	✗	✗	✓	Classif.	(Van Horn et al., 2021)
Places-205	✗	✗	✓	Classif.	(Zhou et al., 2014)
UCF101	✗	✗	✓	Video	(Soomro et al., 2012)
Kinetics-400	✗	✗	✓	Video	(Kay et al., 2017)
SSv2	✗	✗	✓	Video	(Goyal et al., 2017)
GLD v2	✓	✓	✗		(Weyand et al., 2020)
R-Paris	✗	✓	✓	Retrieval	(Radenović et al., 2018a)
R-Oxford	✗	✓	✓	Retrieval	(Radenović et al., 2018a)
Met	✗	✓	✓	Retrieval	(Ypsilantis et al., 2021)
Amstertime	✗	✓	✓	Retrieval	(Yildiz et al., 2022)
ADE20k	✗	✓	✓	Seg.	(Zhou et al., 2017)
Cityscapes	✗	✓	✓	Seg.	(Cordts et al., 2016)
VOC 2012	✗	✓	✓	Seg.	(Everingham et al., 2010)
Mapillary SLS	✓	✗	✗		(Warburg et al., 2020)
NYU-Depth V2	✗	✓	✓	Depth	(Silberman et al., 2012)
KITTI	✗	✓	✓	Depth	(Geiger et al., 2013)
SUN-RGBD	✗	✓	✓	Depth	(Song et al., 2015)
DollarStreet	✗	✗	✓	Fairness	(De Vries et al., 2019)
Casual Conv.	✗	✗	✓	Fairness	(Hazirbas et al., 2021)

Table 18: **List of datasets used.**

Dataset	Pretraining (as is)	Retrieving pretraining data	Eval.	Task	Citation
ImageNet-1k	✗	✓	✓	Classif.	(Russakovsky et al., 2015)
ImageNet-22k	✓	✓	✗		(Deng et al., 2009)
ImageNet-V2	✗	✗	✓	Classif.	(Recht et al., 2019)
ImageNet-ReaL	✗	✗	✓	Classif.	(Beyer et al., 2020)
ImageNet-A	✗	✗	✓	Classif.	(Hendrycks et al., 2021b)
ImageNet-C	✗	✗	✓	Classif.	(Hendrycks & Dietterich, 2019)
ImageNet-R	✗	✗	✓	Classif.	(Hendrycks et al., 2021a)
ImageNet-Sk.	✗	✗	✓	Classif.	(Wang et al., 2019)
Food-101	✗	✓	✓	Classif.	(Bossard et al., 2014)
CIFAR-10	✗	✓	✓	Classif.	(Krizhevsky et al., 2009)
CIFAR-100	✗	✓	✓	Classif.	(Krizhevsky et al., 2009)
SUN397	✗	✓	✓	Classif.	(Xiao et al., 2010)
StanfordCars	✗	✓	✓	Classif.	(Krause et al., 2013)
FGVC-Aircraft	✗	✓	✓	Classif.	(Maji et al., 2013)
VOC 2007	✗	✓	✓	Classif.	(Everingham et al., 2010)
DTD	✗	✓	✓	Classif.	(Cimpoi et al., 2014)
Oxford Pets	✗	✓	✓	Classif.	(Parkhi et al., 2012)
Caltech101	✗	✓	✓	Classif.	(Fei-Fei et al., 2004)
Flowers	✗	✓	✓	Classif.	(Nilsback & Zisserman, 2008)
CUB200	✗	✓	✓	Classif.	(Welinder et al., 2010)
iNaturalist 2018	✗	✗	✓	Classif.	(Van Horn et al., 2018)
iNaturalist 2021	✗	✗	✓	Classif.	(Van Horn et al., 2021)
Places-205	✗	✗	✓	Classif.	(Zhou et al., 2014)
UCF101	✗	✗	✓	Video	(Soomro et al., 2012)
Kinetics-400	✗	✗	✓	Video	(Kay et al., 2017)
SSv2	✗	✗	✓	Video	(Goyal et al., 2017)
GLD v2	✓	✓	✗		(Weyand et al., 2020)
R-Paris	✗	✓	✓	Retrieval	(Radenović et al., 2018a)
R-Oxford	✗	✓	✓	Retrieval	(Radenović et al., 2018a)
Met	✗	✓	✓	Retrieval	(Ypsilantis et al., 2021)
Amstertime	✗	✓	✓	Retrieval	(Yildiz et al., 2022)
ADE20k	✗	✓	✓	Seg.	(Zhou et al., 2017)
Cityscapes	✗	✓	✓	Seg.	(Cordts et al., 2016)
VOC 2012	✗	✓	✓	Seg.	(Everingham et al., 2010)
Mapillary SLS	✓	✗	✗		(Warburg et al., 2020)
NYU-Depth V2	✗	✓	✓	Depth	(Silberman et al., 2012)
KITTI	✗	✓	✓	Depth	(Geiger et al., 2013)
SUN-RGBD	✗	✓	✓	Depth	(Song et al., 2015)
DollarStreet	✗	✗	✓	Fairness	(De Vries et al., 2019)
Casual Conv.	✗	✗	✓	Fairness	(Hazirbas et al., 2021)

表18：所用数据集列表。